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Is amblyopia reflected in spontaneous EEG alpha rhythms?

Gabriel Werpel Fernandes

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The University of Auckland

10 November 2024

Supervisor: Dr Luke E. Hallum

A dissertation submitted for partial fulfillment of the requirements for the MECHENG.795A/B project in Mechanical Engineering, The University of Auckland, 2024.

ABSTRACT

Amblyopia, a visual disorder affecting 1-5% of the population, is defined by reduced vision in one eye and is associated with alterations in neural processing within the visual cortex. Alpha rhythms (8-12 Hz) in electroencephalogram (EEG) recordings are indicative of relaxed wakefulness and are linked to cognitive and perceptual functions, with the primary visual cortex (V1) being a significant source of these oscillations. This study aims to investigate whether amblyopia causes distinct changes in EEG signals, particularly focusing on alpha rhythms, and to evaluate the efficacy of machine learning (ML) classifiers in distinguishing amblyopic patients from individuals with normal vision based on these neural signatures. Such classifiers could be useful in diagnosis and assessment of amblyopia severity. EEG data were collected from individuals diagnosed with amblyopia and control subjects with normal vision across various states of eye closure: both eyes closed, both eyes open, dominant eye closed, and non-dominant eye closed. Frequency-domain features, specifically Fourier coefficients, were extracted from the EEG signals. ML techniques, including Extreme Gradient Boosting (XGB), were employed to classify participants as amblyopic patients or control subjects based on their EEG data. Feature importance was analysed, and the robustness of the findings was validated using permutation tests to ensure that the classification performance was not attributable to random chance. XGB achieved the best score with an accuracy of 85% (chance performance = 50%) in distinguishing between amblyopic patients and control subjects, demonstrating its proficiency in handling complex EEG data. Contrary to the initial hypothesis, significant increases in alpha activity were not observed when only the dominant eye was closed in amblyopic individuals. Instead, the Fourier coefficients effectively captured neural oscillatory differences across multiple frequency bands and conditions, including theta (4–8 Hz), alpha (8–12 Hz), beta (16–30 Hz), and high-gamma (36–40 Hz). Notably, the combination of amplitude ratio features from the central occipital electrode (Oz) under the dominant eye closed condition and the left occipital electrode (O1) under the non-dominant eye closed condition yielded a classification accuracy of 83% with a Matthews correlation coefficient (MCC) of 0.65 (chance performance, MCC = 0.0). The findings suggest that amblyopia affects a broader range of neural oscillatory activities beyond the alpha band, particularly within the occipital regions of the brain. The high classification accuracy achieved by ML models highlights the potential of EEG-based biomarkers for non-invasive amblyopia diagnosis. However, limitations such as data artifacts and a limited sample size underscore the need for further research with larger, more diverse populations and enhanced data preprocessing techniques. Future studies should aim to validate these preliminary findings and explore the integration of additional EEG features and advanced classifiers to develop reliable diagnostic tools for amblyopia.

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Acknowledgements

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Abbreviations

AD	Alzheimer's Disease
ANOVA	Analysis of Variance
AR	Autoregressive
ARI	Alpha Reactivity Index
BDF	Binary Data Format
COH	Coherence of Peak
CV	Cross Validation
DBM	Digital Brain Mapping
DEC	Dominant Eye Closed
EC	Both Eyes Closed
EEG	Electroencephalogram
EO	Both Eyes Open
ETDRS	Early Treatment Diabetic Retinopathy Study
FFT	Fast Fourier Transform
fMRI	Functional Magnetic Resonance Imaging
FN	False Negative
FP	False Positive
GB	Gradient Boosting
HC	Healthy Control
IQ	Intelligence Quotient
LC	Left Eye Closed
LGN	Lateral Geniculate Nucleus
LOO	Leave-One-Out
LR	Logistic Regression
LS	Light Stimulation
LTH	Light Therapy
MCC	Matthews Correlation Coefficient
MCI	Mild Cognitive Impairment
ML	Machine Learning
NaN	Not a Number
NDEC	Non-Dominant Eye Closed
NN	Neural Network
O1	Left Occipital Electrode
O2	Right Occipital Electrode

OCTA	Optical Coherence Tomography Angiography
Oz	Central Occipital Electrode
PET	Positron Emission Tomography
PSD	Power Spectral Density
qEEG	Quantitative Electroencephalography
RAS	Reticular Activating System
RC	Right Eye Closed
RDR	Rhythmic Driving Response
RF	Random Forest
SA	Strabismus and Amblyopia
SED	Sensory Eye Dominance
SSVEPs	Steady-State Visual Evoked Potentials
STFT	Short-Time Fourier Transform
SVM	Support Vector Machine
TN	True Negative
TP	True Positive
V1	Primary Visual Cortex
VA	Visual Acuity
VEPs	Visually Evoked Potentials
WMH	White Matter Hyperintensities
WT	Wavelet Transform
XGB	Extreme Gradient Boosting / XGBoost

Nomenclature

Units of Measurement

dB	Decibels
Hz	Hertz
V	Volts

Greek Symbols

μ	Micro, Mean
σ	Standard Deviation
σ^2	Variance

1 Introduction

1.1 Background

1.1.1 Amblyopia

Amblyopia, also known as "lazy eye," is a neurodevelopmental disorder defined by reduced vision in one eye. The causes of amblyopia include:

- strabismus (misalignment of the eyes)
- anisometropia (significant differences in refractive error between the two eyes)
- stimulus deprivation (blockage of vision in one eye due to cataracts)

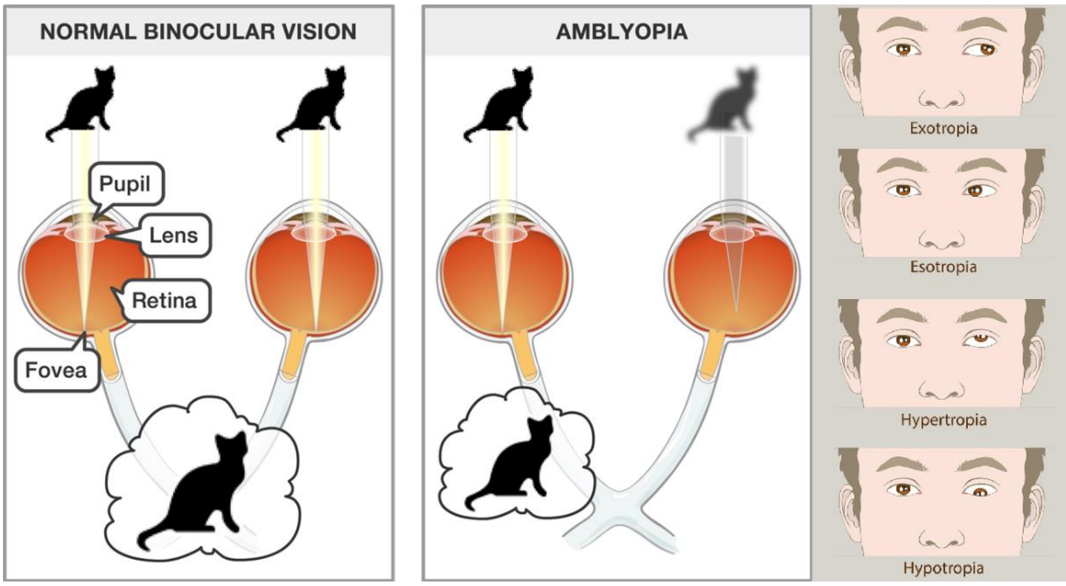

Figure 1 Visual comparison between normal binocular vision and amblyopia [20]. Examples of strabismus [21].

Its condition arises from abnormal visual information processing, generally within the first few years of life, up to age 7. Within this period, the visual system is highly neuroplastic and capable of development, and the brain's ability to adapt is high. However, if left untreated, the changes in visual pathway development can become permanent, highlighting the importance of early detection and intervention [1]-[4].

The implications of untreated amblyopia extend beyond visual acuity. They affect binocular vision, which plays a critical role in some forms of depth perception, as well as the ability to perceive the world in three dimensions. In addition, individuals with untreated amblyopia often experience social and psychological effects, including decreased quality of life, academic challenges, and reduced employment opportunities [5].

The significance of studying amblyopia lies in its prevalence, affecting approximately 1-5% of the global population, and its potentially reversible nature if detected and treated early. Amblyopia is classified as the most common cause of monocular visual impairment among children and young adults, making it a significant public health concern [1]-[3].

Diagnostics for amblyopia involve visual assessment, including visual acuity tests that measure the clarity of vision at a certain distance, binocular function tests to assess the ability of the eyes to work together, also the use of imaging techniques such as Optical Coherence Tomography Angiography (OCTA). In addition, the literature highlights the potential of diagnostic processes involving electroencephalography-based methods, such as VEPs (Visually Evoked Potentials) and entropy estimation, which provides non-invasive ways to assess the neural mechanisms affected by amblyopia [6]-[10].

Subjective and objective methods play distinct roles in diagnosing amblyopia, each with its advantages and limitations. Subjective methods, such as letter acuity tests, primarily rely on the patient's responses to visual stimuli. These tests are widely used due to their simplicity and directness in assessing visual clarity and sharpness. However, they can be less reliable because they depend on patient cooperation and interpretation, which may vary, especially in young children or in distinguishing different types of amblyopia. On the other hand, objective methods, including VEPs and entropy estimation, do not require patient input about what they see. These techniques assess the electrical activity in the brain in response to visual stimuli, offering a more direct measurement of the visual system's functioning. They provide valuable insights into the neural mechanisms affected by amblyopia, thereby potentially increasing diagnostic accuracy. These methods can specifically identify abnormalities in brain function that subjective tests might miss, thus offering a more robust framework for distinguishing between different forms of amblyopia and understanding its neurophysiological impacts.

1.1.2 Electroencephalogram

EEG is a technique that measures brain activity via the brain's electrical oscillations. It is used in research and clinical diagnostics to evaluate brain functions, diagnose neurological disorders, and study the brain's electrical activities during cognitive processes, such as visual perception [11].

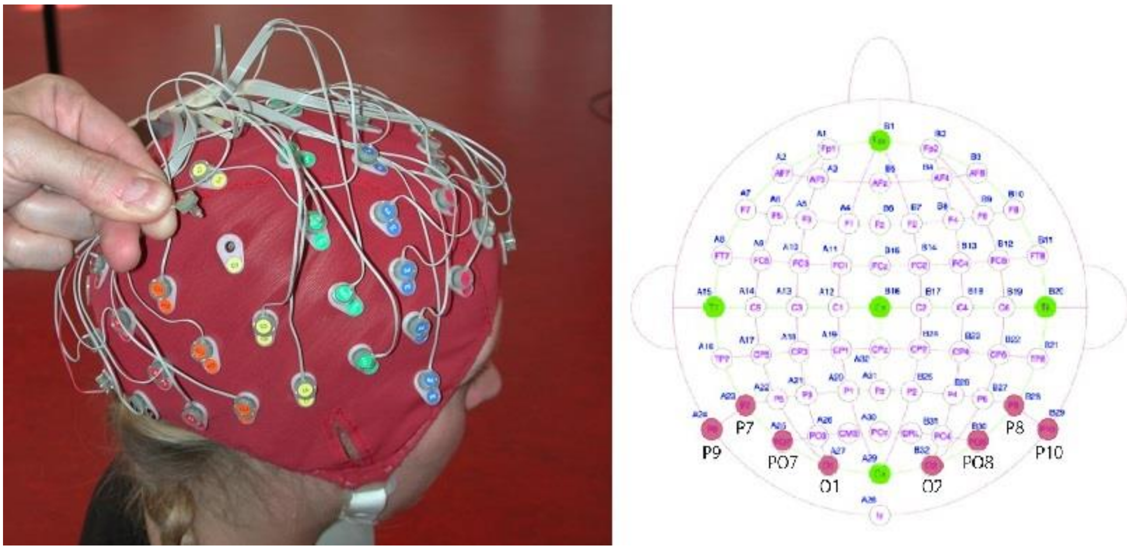

Figure 2 Biosemi EEG device used for reading brain electrical activities [22]. Electrodes layout with red-dot channels situated in the parietal-occipital area [23].

EEG's non-invasive nature is advantageous. It allows repeated measurements over time without exposing patients to the risks of more invasive methods. This advantage is crucial for longitudinal studies, to understand the progression of visual impairments and evaluate the neuroplastic changes following interventions. Moreover, EEG's ability to detect alterations in brain activity that may precede observable symptoms offers the potential for early diagnosis, significantly impacting effective treatments [12]-[14].

Now, with the increase of computational power, researchers have finally begun to understand the complex neural mechanisms from conditions such as amblyopia by analysing specific brain wave patterns. The studies typically focus on the alterations in EEG waveforms, captured from the parietal-occipital areas, associated with visual stimulation or the lack of it, aiming to identify characteristic biomarkers that can inform both diagnosis and the effectiveness of therapeutic interventions [15].

Using EEG in studies and treating brain and eye disorders is a big step towards personalized healthcare. As this technology improves, its role in understanding brain functions and disorders will only grow since it provides more accurate and easier use. Future research using advanced EEG systems and complex data analysis will help us learn more about the brain processes involved in visual disorders, leading to less invasive treatments.

1.1.3 Primary Visual Cortex and Alpha Rhythms

Building on the fundamental role of EEG in diagnosing and studying visual and neurological disorders, it is important to understand specific brain structures and rhythms, particularly the V1 and alpha rhythms.

The V1, situated in the posterior part of the brain's occipital lobe, is essential for processing visual information. Since V1 is the endpoint for visual input coming from the retina to the brain, its functionality is critical in interpreting aspects of visual perception such as colour, orientation, and motion [16]. Moreover, regarding the V1's specific role, it not only processes visual information but also contributes to the generation and modulation of alpha rhythms., which are notably observed in EEG recordings [32], [35].

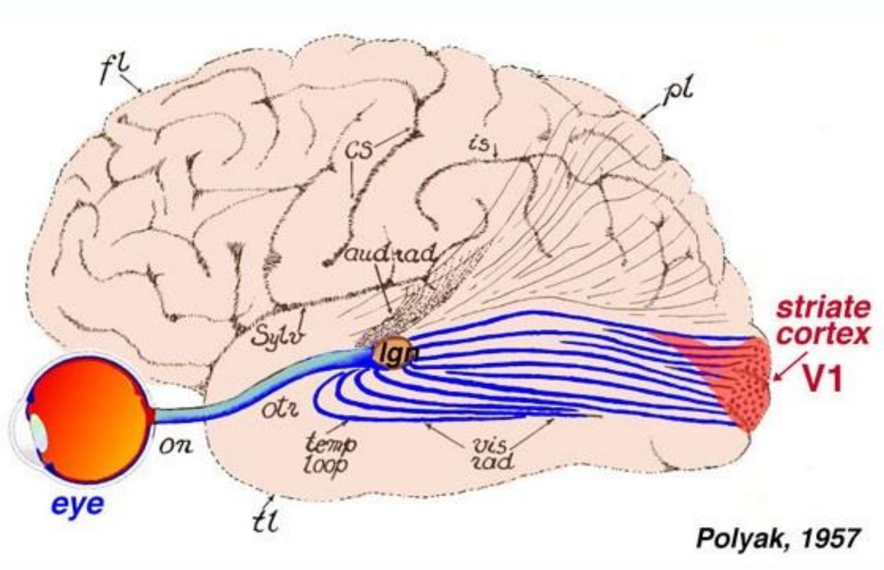

Figure 3 Visual information travels from the eye to the Lateral Geniculate Nucleus (LGN) and reaches the primary visual cortex, also known as V1, at the back of the occipital lobe [17].

Research suggests that alpha rhythms are actively involved in gating sensory processing and attention. The alpha activity in V1 can be influenced by visual stimuli, attention, and other cognitive processes, changing the information flow within the visual cortex and across other cortical areas [35]-[36].

Alpha rhythms, defined by their 8-12 Hz frequency band, are most prevalent during states of quiet wakefulness, particularly when the eyes are closed but also when they are open in a resting state. These rhythms are indicative of an inhibitory process that modulates sensory input and cognitive load, and are associated with various cognitive processes, including attention, learning, and memory. The presence and modulation of these rhythms provide insights into the brain's state of alertness and rest [18].

In V1, alpha rhythms may act to inhibit the processing of non-essential information during states of relaxed focus, helping to conserve resources for more significant processing tasks. This role of alpha rhythms in gating and prioritising information is integral to how the brain manages sensory input and cognitive load [35],[37].

The generation of spontaneous alpha waves, particularly in the V1, is closely linked to the thalamocortical system. In this context, the reticular activating system (RAS) plays an important role. The RAS is a network of neurons in the brainstem, responsible for sensation, consciousness, attention, and the sleep-wake cycle. It influences cortical activity through its connections with the thalamus, which is a relay station for sensory information to the cortex. The RAS adjusts the flow of sensory inputs relayed by the thalamus to the cortex, a modulation that controls the cortical arousal state and influence the generation of alpha rhythms [33]-[34].

In essence, V1 contributes to the generation and modulation of alpha rhythms by interacting with the thalamic relay influenced by the RAS. This interaction helps regulate cortical excitability and information flow, which are essential for visual processing and cognitive functions involving attention and sensory gating.

The relationship between spontaneous alpha waves and visual processing is factual. During visual tasks, the amplitude of alpha waves typically decreases, reflecting the cortical activation necessary for processing visual stimuli. Conversely, increased alpha wave activity indicates a state of cortical inhibition, often observed in the absence of visual stimulation [19].

Understanding the intricacies of how the primary visual cortex and alpha rhythm's function and interact provides insights into the neural mechanisms underlying visual processing and disorders such as amblyopia. By studying V1 and alpha rhythms, researchers can further enhance diagnostic techniques and therapeutic approaches, leveraging the non-invasive and informative nature of EEG.

1.2 Literature Review

1.2.1 Alpha Rhythms and EEG Analysis in Neurological Research

Research comparing alpha rhythms across different contexts and populations has a longstanding history in the field of EEG studies. It has been proved that the power spectral density (PSD) of alpha in the occipital region decreases when transitioning from a resting state with eyes-closed to a resting state with eyes-open [28].

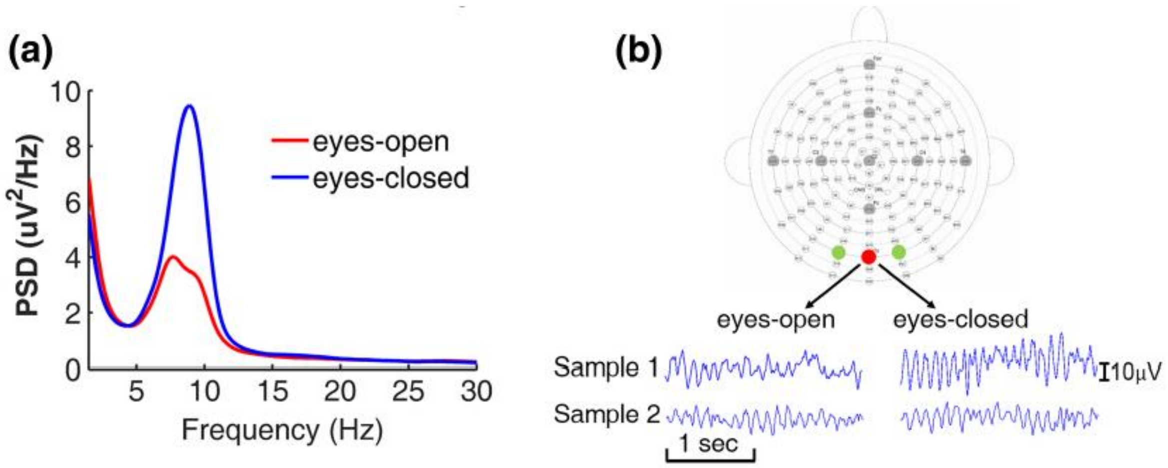

Figure 4 Alpha oscillations during both eyes-closed (EC) and both eyes-open (EO) resting state in older adults. (a) Power spectra from occipital channels averaged across participants. (b) Two participants showing strong (sample 1) and weak (sample 2) alpha reactivity [30].

A research about this phenomenon involves examining alpha rhythms during different developmental stages. Smith [29] tries to identify and describe the EEG patterns across different age groups, providing a reference for EEG development. A cross-sectional approach was used and a cohort of children and adults ranging in age from 0 to 32 years. The EEG was analysed for frequency, amplitude, and waveform characteristics. While exact methodological details were not explicitly outlined, it's inferred that the EEGs were conducted under standardised conditions for specific brain regions. The study found that the alpha waves vary across different ages. For instance, there is a trend indicating that as children grow older, specific frequency bands become more dominant. In older children (10+ years old), alpha rhythms became more prominent. It was also noted that the alpha frequency waves are directly proportional to the metabolic activity. A rank difference correlation of 0.9 suggests that this controlling factor might determine the differences in variations between children and adults. Smith's study concluded that EEG patterns develop systematically as children age, with each age group showing distinct waveform characteristics. The progression of dominant rhythms and changes in their frequencies provides a framework to distinguish normal EEG development. Even though the paper provides conclusive results, it has a limited sample size and lack demographic diversity, which reduces the generalisability of its findings. Additionally, EEG recording equipment from the 1930s did not have the same resolution or consistency as modern tools, potentially affecting data fidelity. Furthermore, the study does not specify the exact conditions under which the EEG was recorded, such as the state of wakefulness or electrode placement, both of which could influence results. At last, modern statistical analysis or signal processing methods could provide more accurate interpretations of the recorded signals.

Another approach that leverages this knowledge of alpha rhythms is to explore its association with neurological diseases, thereby enhancing our understanding of these conditions. A study conducted by Chae et al. [31] examined whether alpha reactivity from EC-to-EO conditions could serve as an early biomarker for cerebral amyloid-β (Aβ) deposition in non-demented older adults. The study involved 143 non-demented patients aged ≥55 years who visited a memory clinic. These participants underwent a combined EEG and [18F] florbetaben Positron Emission Tomography (PET) study to assess Aβ deposition and were divided into Aβ-positive (Aβ+) and Aβ-negative (Aβ-) groups. EEG recordings were taken during EC and EO conditions, and a Fourier transform spectral analysis was performed. The analysis focused on alpha-band power and included various covariates such as age, sex, education, global cognition, medication use, depression, and white matter hyperintensities (WMH). The study found a significant 3-way interaction among Aβ positivity, eye condition, and laterality factor on alpha-band power. The EC-to-EO alpha reactivity in the left hemisphere was reduced in Aβ+ subjects without dementia. The alpha reactivity index (ARI) was lower in the left hemisphere of Aβ+ subjects and was more pronounced in the left posterior area. The study concluded that reduced EC-to-EO alpha reactivity, particularly in the left hemisphere, acts as an early biomarker for cerebral Aβ deposition. Chae's study provides insights into the potential of alpha reactivity as an early biomarker for Alzheimer's disease (AD). One of its strengths lies in the comprehensive evaluation, which included extensive clinical, neuropsychological, and neuroimaging assessments, resulting in a robust dataset. Additionally, the adjustment for various covariates affecting brain electrical activities improved the reliability of the findings. However, the study also has notable limitations. The sample size, although relatively large, predominantly included individuals with mild cognitive impairment (MCI), limiting the generalisability of the findings to a broader population. The topographic analysis, conducted with a 19-channel EEG recording, was somewhat crude, and more detailed analyses could yield additional insights. To improve future research, larger and more diverse samples, including individuals with preclinical AD should be considered. Employing more advanced EEG analysis techniques and higher-resolution EEG recordings could provide deeper insights into alpha reactivity changes.

Longitudinal studies would help determine the consistency and reliability of alpha reactivity as a marker throughout AD progression.

It must be mentioned that a critical aspect of EEG research is establishing normative data for comparison, which is characterised by what is usual within a population at a specific point or period of time. The study from Wieneke et al. [15] aimed to obtain objective normative data about the properties of the alpha rhythm in EEGs. The study involved 110 male subjects, specifically airplane pilots, selected for their routine EEG examinations. The subjects were divided into age groups, three years range each, from 18 to 50 years, with ten subjects in each group. EEG recordings were performed using the 10-20 system with eight electrodes placed at T3, T5, P3, O1 on the left side and T4, T6, P4, O2 on the right side, referenced to Fp1 and Fp2. The EEG data, recorded by a 16-channel apparatus, underwent spectral analysis using Fast Fourier Transform (FFT) over periods with eyes open and eyes closed. To enhance the clarity and utility of the spectral data, spectral convolution was also used. This method smoothed the spectra, thereby providing a clearer view of significant peaks which are crucial for identifying typical alpha rhythm patterns. The analysis included statistical calculations such as means, standard deviations, and confidence intervals to describe the distribution and variability of these EEG parameters among the normal population. Parameters such as peak frequency, power density, bandwidth, reactivity, and coherency functions were calculated for each experiment.

- *Frequency*: The alpha rhythm peak frequencies were reported to have a mean of 9.9 Hz with a standard deviation of 1.0 Hz, and the distribution of these frequencies was slightly skewed.

- *Peak Power Density*: Measured in dB, power density reflects the intensity of brain activity within the alpha band, providing a quantifiable measure of brain rhythm strength. Alpha peak power density showed considerable inter-individual variability, with values being plotted logarithmically to emphasize differences across the range more clearly. The study also observed an interhemispheric asymmetry in alpha power, with the right hemisphere generally exhibiting higher power than the left, consistent with existing literature [31].

- *Reactivity*: defined as the difference in power density between eyes open and eyes closed. No significant conclusions were taken within the reactivities experiments.

- *Bandwidth*: The mean bandwidth of the alpha rhythm was reported as 1.30 Hz with a standard deviation of 0.55 Hz. This parameter indicates the frequency range around the peak frequency where significant power is maintained.

- *Coherency*: Inter-hemispheric coherency of alpha rhythms, analysed using Fisher's z-transformed values, varied among individuals, highlighting differences in the synchronization of neural activity across the brain. Alpha coherency was highest in the occipital regions and decreased towards the temporal regions, with postero-anterior coherency surpassing left-right coherency in the occipital areas.

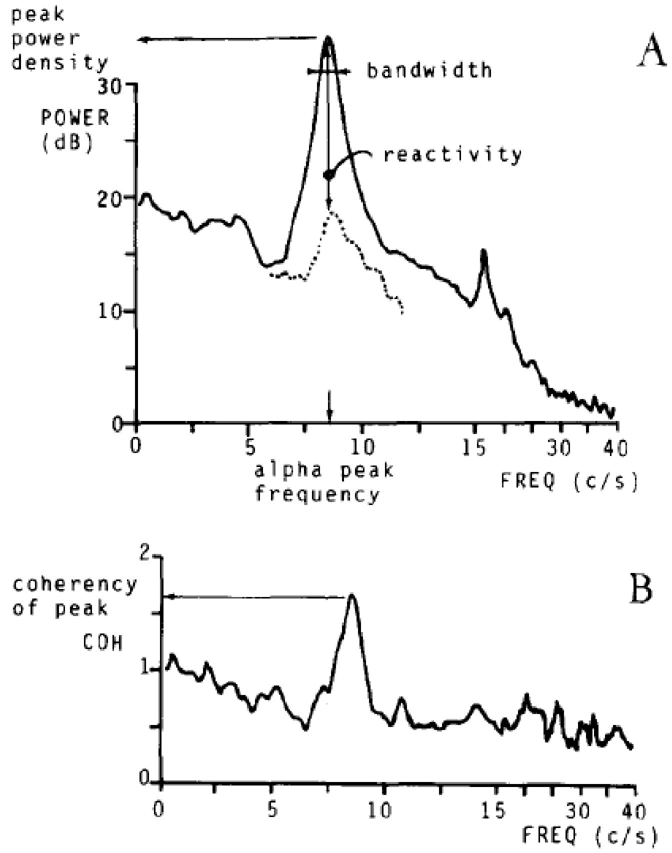

Figure 5 A: an example of a spectrum with a clear alpha peak with eyes closed (continuous line). The spectrum with eyes open is drawn as a dotted line only near the alpha peak. The spectral parameters which were determined are indicated. B: an example of the coherency function of the EEGs of two derivations with eyes closed. The coherency parameter is indicated [15].

The data on interhemispheric asymmetry and coherency further contribute to understanding the spatial distribution and synchronization of alpha rhythms. These normative values are good for clinical applications, offering a reference point for distinguishing normal from pathological EEG patterns. The study emphasizes the utility of quantitative spectral analysis in clinical neurophysiology, suggesting that these parameters can aid in the accurate diagnosis of neurological conditions. Despite the comprehensive article from Wieneke's, the study has areas for improvement. The exclusive focus on male airplane pilots may introduce selection bias, limiting the generalisability of the findings to the broader population, including females, infants and individuals from different occupational backgrounds. Additionally, the study's reliance on technology from the 1970s, such as analog tape for the recordings, with its constraints on effective bandwidth and epoch duration, may limit the detection of finer spectral details. The study could also benefit from addressing biases in power density and coherency calculations more rigorously, possibly through advanced statistical methods.

The study by Zhang et al. [27] serves as an excellent resource for understanding the various methodologies used in EEG signal analysis, their applications, and their implications in both research and clinical contexts. The paper highlights strengths and weaknesses of different EEG analysis methods. For instance, power spectrum analysis methods like FFT and Welch are useful for identifying frequency components but may be affected by noise. Time-frequency methods like Short-Time Fourier Transform (STFT) and Wavelet Transform (WT) provide detailed temporal and spectral information but may suffer from resolution trade-offs. Connectivity analysis methods help understand functional relationships between brain regions, while source localisation methods pinpoint the origin of EEG signals. ML approaches offer advanced pattern recognition capabilities but require large datasets and computational resources. While the paper provides a thorough review of EEG analysis methods, there are several areas where it could be improved. The paper could benefit from a more detailed comparison of methods in terms of their performance metrics, computational requirements, and suitability for different types of EEG data. Including case studies or practical examples of how these methods have been successfully implemented in research or clinical practice would enhance the practical utility of the paper. The study could be expanded to include more recent and emerging EEG analysis techniques, such as deep learning approaches and advanced source reconstruction methods. Discussing the integration of EEG with other neuroimaging modalities, such as Functional Magnetic Resonance Imaging (fMRI) or Magnetoencephalography (MEG), could provide a more holistic view of brain function analysis. The authors conclude that no single EEG analysis method is universally superior. The choice of method depends on the specific research objectives and characteristics of the EEG data.

In neurological research, the exploration of alpha rhythms and EEG analysis has proven to be a significant area of study. Our research on amblyopia and alpha rhythms using EEG, inspired by the methods discussed in Wieneke's study and Zhang's article, is proof to this. Power spectrum analysis may help identify differences in frequency components between amblyopic and normal subjects. Which leads us to the question that forms the basis of our discussion: "Is amblyopia reflected in spontaneous EEG alpha rhythms?". We aim to uncover the relationship between amblyopia and EEG alpha rhythms, thereby shedding light on the potential of EEG as a diagnostic tool for this specific disorder.

1.2.2 EEG rhythms in amblyopic patients

Studies have shown that amblyopia can lead to alterations in EEG rhythms, reflecting the disruptions in cortical plasticity and neural connectivity associated with amblyopia. Here we revise the current literature on EEG rhythms in amblyopic patients, focusing on how neural oscillations are affected by the disorder and their implications for understanding the cortical deficits in amblyopia.

Ibrahimi et al. [38] assesses the brain activity and visual performance of patients with strabismus and amblyopia (SA) before and after a complete cycle of light therapy (LTH).

Figure 6 Light delivery to achieve photo biomodulation therapy. Example of LTH using transcranial approach [44].

The study aimed to determine whether LTH could enhance brain synchrony and visual performance in these patients using quantitative electroencephalogram (qEEG) analysis. The study included 17 patients with SA and 11 healthy controls (HCs). The inclusion criteria for SA patients were primary strabismus and amblyopia, visual acuity of ≥0.7 logMAR, age 8–30 years, and normal intelligence quotient (IQ). Exclusion criteria included secondary strabismus, previous eye surgeries, photosensitivity, and certain medical conditions. Participants underwent medical history collection, visual performance evaluations, EEG recordings, and a 20-session LTH program. The qEEG was used to measure brain activity at baseline and after LTH. The data were analysed using mixed Analysis of Variance (ANOVA) for repeated measures, Spearman correlation, and other statistical tests to compare visual performance and brain activity between SA patients and HCs at baseline and post-LTH. The results of the brain activity study showed that the alpha wave frequency was higher in HCs compared to SA patients at both baseline and post-Low LTH. This difference was found to be statistically significant for the groups ($p < 0.001$). The alpha wave values increased after LTH in both groups but more significantly in HCs. The paper concludes that LTH induced positive changes in both brain activity and visual performance in SA patients. This was evidenced by a significant increase in alpha wave frequency, enhanced interhemispheric synchronicity, and improved alpha wave distribution. The paper's generalisability is limited due to the small sample size, particularly in the HC group. The study did not evaluate the long-term effects of LTH, and follow-up assessments could've offerred insight into observed changes. The paper refers to previous work for specific technical details about the qEEG analysis using Neuronic, leaving readers without direct access to crucial methodological information within the current paper. The terminology could be clarified by using "sensory dominance" to describe the predominant eye used for visual processing, aligning better with standard terminology in the field. The paper hypothesizes that LTH influence the alpha wave distribution, anteroposterior gradient, and interhemispheric synchronicity, but lacks a clear rationale for why these specific measures are affected. Details on how alpha wave frequency was measured are missing, and the context of the EEG recordings (e.g., what subjects were doing) is unspecified. The terms "low voltage" and "high voltage" are used without clear definitions or explanations of how these measures were derived. The paper could further elaborate on the implications of the correlation analysis beyond test-retest reliability, discussing how the correlations inform the stability and changes in brain activity patterns due to LTH. The spatial distribution of alpha waves lacks clarity on how this distribution was quantified. The author could include the metrics or algorithms used to assess coherence between hemispheres. The statement attributing irregular alpha wave distribution to the hemisphere in charge of visual processing is incorrect, as literature [40] confirms that both hemispheres receive input from each eye. The author could also clarify how cortical adaptations may influence EEG patterns. In addition, the choice of statistical methods for different analyses requires justification. The author should ensure consistency and provide reasons for selecting specific tests, such as ANOVA, to analyse visual performance data.

Another study related to EEG rhythms in amblyopic patients from Ibrahimi et al [41] measures and compare the baseline cortical activity during light stimulation (LS) in patients with SA and HCs. The aim was to understand the differences in brain functionality and propose LS as a potential brain stimulator for these patients. This observational, longitudinal, prospective study involved 17 SA patients and 17 HCs. The methods included EEG and digital brain mapping (DBM) to identify changes in frequency, voltage, and brain coherence. EEG recordings were taken at baseline and during a 20-minute LS session using filters transmitting blue and red light. The study used both parametric and non-parametric statistical analyses (Mann-Whitney U test, paired t-test, Wilcoxon test) to compare EEG measurements between the groups and within each group. The baseline findings of the study suggest that initially, patients diagnosed with SA exhibit a decreased frequency of alpha wave activity ($p = 0.029$), which is distributed abnormally across the hemispheres. In the case of the HC group, LS modifies the distribution of alpha waves across the hemispheres and affects the condition of synchronicity between the hemispheres. No significant alterations were observed in the frequency of the alpha wave or the anteroposterior gradient. The author concludes that LS is suggested as a potential brain stimulator and a complementary therapy for treating patients with SA. Once again, a sample size increase from Ibrahimi would provide more robust and generalisable conclusions. Additionally, extending the duration and frequency of LS sessions in this longitudinal study could offer insights into the long-term effects of LS. Including a more diverse population would help understand the broader applicability of LS across different demographics. Implementing more rigorous control of variables, such as participants' sleep patterns, diet, and other lifestyle factors, would improve the study's reliability.

Chen et al. [42] study was aimed to evaluate the suppression of refractive amblyopia and understand the neural mechanisms underlying amblyopia suppression, also the efficacy of push-pull perception training. The methods involved recording EEG of children with refractive amblyopia before, during, and after push-pull perception training, comparing brain activity in different states through steady-state visual evoked potentials (SSVEPs) and power topography, and comparing these with normal children.

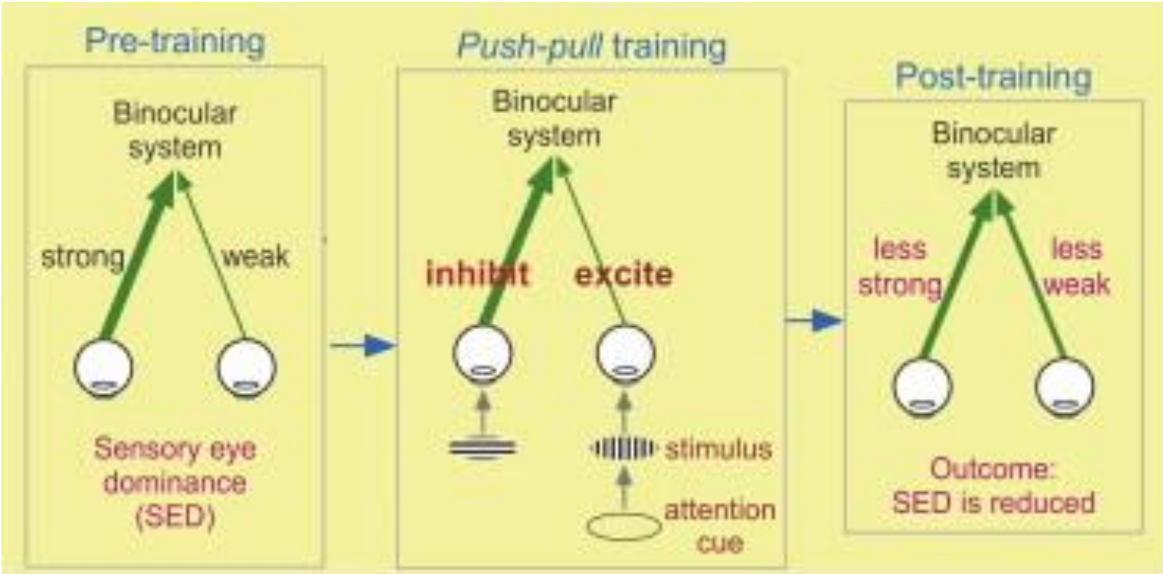

Figure 7 Push-Pull Perception Training [45].

The results related to alpha rhythms indicated that push-pull perception training led to a reduction in alpha power in the occipital and temporal lobes. During SSVEP response before training, the fundamental frequency response at 7 Hz was significantly weaker in amblyopic eyes compared to fellow eyes. The response in amblyopic eyes was lower, indicating reduced visual cortex activity in processing visual stimuli at this frequency. After the SSVEP response training, the post-training measurements showed no significant differences at 7 Hz between amblyopic and fellow eyes, suggesting that the training helped equalize visual processing capabilities between the two eyes. The change in the intermodulation response and the alpha power in the occipital lobe were significantly correlated with clinical indicators of visual function. The conclusions highlighted that EEG is a promising method for measuring amblyopia suppression and the effectiveness of push-pull perception training. Specifically, the training improved SSVEP performance, reduced differences in SSVEP response between eyes, and improved intermodulation frequency response. However, potential limitations include the small sample size and the focus primarily on amblyopic children. The study's reliance on EEG measurements also suggests potential variability in data quality due to subject cooperation or movement artifacts. Improvements could include integration of additional neuroimaging tools to validate and extend the EEG findings.

A recent study from Boichuk et al. [43] identifies the features of the visual system state and changes in the background EEG in response to activation procedures in SA depending on the type of fixation in the amblyopic eye. The study involved 22 strabismic amblyopic patients aged 5-8 years and 15 healthy children, around the same age. EEGs were performed using the 10–20 system of electrode placement, and the background EEG rhythms, interhemispheric asymmetry, and alpha wave percentages were assessed. Potentials in the brain were recorded using standard activation procedures to activate deep brain structures.

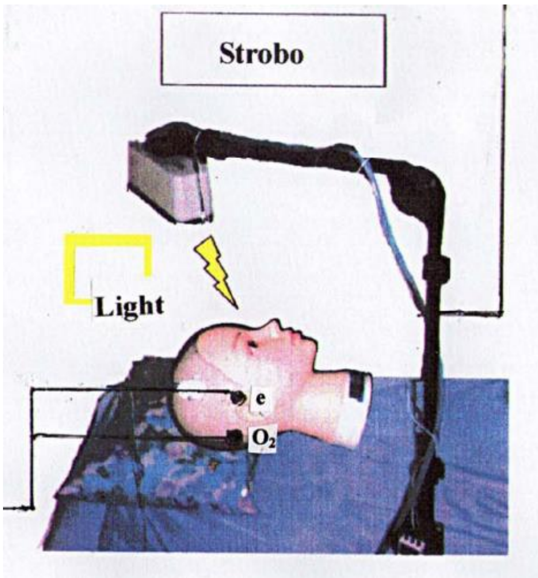

Figure 8 Setup example of eye opening and photic stimulation [46].

This study employed statistical methods to analyse the results, including the Kolmogorov-Smirnov test and Student t-test. The results indicated that the alpha rhythm was not discernible in the background EEG in the occipital region in 3.0 ± 2.6% of cases, and the alpha index was below the normal range in 48.6 ± 8.2% of patients with a visual acuity worse than 0.3. Reduced alpha activity was found in almost half of the strabismic amblyopic children, indicating immaturity of the synchronizing system and impaired oculomotor tuning. Abnormal bilateral EEG responses to eye-opening stimulation and abnormal bilateral EEG rhythmic driving response (RDR) were observed in most children with eccentric fixation, but not in those with central fixation. No definitive conclusion could be drawn regarding the alpha waves. From this study, increased theta and delta wave percentages in frontal and occipital derivations were found in SA compared to HC. Abnormal EEG responses to eye-opening and RDR were prevalent, indicating functional changes in the midline brain structures, including the corpus callosum, which impede the development of binocular vision in SA’s. This study could employ more advanced EEG analysis techniques to gain deeper insights into the neurophysiological mechanisms underlying strabismic amblyopia. They could also explore the longitudinal effects of therapeutic interventions on EEG patterns in amblyopic patients.

1.2.3 Consensus in Amblyopia Research via EEG

1. *Alpha Wave Activity*:

- *General Findings*: There is a consensus that SA is associated with alterations in alpha-wave activity. This manifests as reduced alpha wave frequency and power in patients compared to HCs.

- *Specific Observations*:
    - Ibrahimi et al. [38] & [41]: Studies demonstrated that alpha wave frequency is initially lower in patients with SA than in HCs. Interventions such as LTH and LS not only increased alpha wave frequency but also improved synchronicity, with more pronounced effects in HCs.
    - Chen et al. [42]: Push-pull perception training was found to reduce alpha power in the occipital and temporal lobes, indicating that this training can enhance visual processing abilities.

2. *Therapeutic Interventions*:

- *General Effectiveness*: There is broad agreement on the positive impacts of therapeutic interventions on EEG patterns and visual performance in amblyopic patients.
- *Specific Findings*:
    - Ibrahimi et al. [38] & [41]: Both LTH and LS were effective in enhancing brain synchrony and visual performance in SA patients, suggesting these could be viable treatments for improving cortical deficits in amblyopia.
    - Chen et al. [42]: Push-pull perception training not only improved visual processing capabilities but also equalised the SSVEP responses between the amblyopic and fellow eyes, showcasing its potential as an effective therapeutic tool.

3. *Interhemispheric Synchrony*:

- *Challenges in Synchronicity*: Abnormalities in interhemispheric synchrony appear to be a significant issue in amblyopic patients, indicating disrupted cortical connectivity.
- *Specific Insights*:
    - Ibrahimi et al. [38]: LTH led to increased interhemispheric synchronicity in SA patients, suggesting a potential pathway for therapeutic intervention.
    - Boichuk et al. [42]: Found abnormal interhemispheric asymmetry and reduced alpha activity, indicating impaired cortical synchrony which could be a target for corrective interventions.

These points integrate findings from studies, which presents results of the alterations in EEG rhythms associated with amblyopia.

1.2.4 Gaps in Amblyopia Research via EEG

Current research efforts to find biomarkers for neurological conditions have benefitted from modern statistical methods and advancements in technology. These developments have not only reduced the computational power required but have also enhanced the reliability of data analysis techniques. This constant progress allows researchers to uncover biomarkers that were previously difficult to detect using older methodologies.

Despite advancements, several critical gaps persist in the field:

- *Methodological Diversity*: Many studies fail to employ multiple analytical methods to corroborate their findings, which could strengthen the validity of observed behaviours.

- *Normative Data*: There is a notable deficiency in the literature regarding well-established normative data across the time domain, frequency domain, and time-frequency domain. Such data are essential for standardising analysis methods and aiding in the accurate diagnosis of conditions.
- *Practical Implementation*: Few studies provide practical examples or case studies on the application of these methods in real-world clinical or research settings, which limits understanding of their practical utility.
- *Diagnosis Techniques*: Other methods for diagnosing amblyopia remain underdeveloped, with most recent studies focusing primarily on responses to stimuli rather than analysing spontaneous alpha rhythms.

Some of these methods can enhance the understanding and treatment of amblyopia:

- *Longitudinal Studies*: More long-term studies are needed to evaluate the durability of therapeutic effects observed in short-term interventions.
- *Methodological Details*: Studies should include detailed descriptions of EEG recording conditions and analytical techniques to improve reproducibility and generalisability.
- *Larger and Diverse Samples*: Research often suffers from small sample sizes and limited demographic variability. Expanding these aspects could improve the generalisability of findings.
- *Integration of Neuroimaging Tools*: Integrating additional neuroimaging modalities such as fMRI or PET could help corroborate EEG findings and provide deeper insights into neural mechanisms.
- *Mechanistic Explanations*: There is a need for clearer explanations of how EEG changes relate to neurophysiological and functional outcomes.
- *Control of Variables*: More rigorous control of external variables like lifestyle factors (e.g., sleep, diet) could enhance the reliability of research findings.

By addressing some of these gaps, researchers can develop more effective diagnostic and therapeutic strategies for amblyopia, contributing to a better understanding of its underlying neural mechanisms and long-term treatment outcomes.

2 Scope and Objectives

2.1 Scope

1. *Data Collection*: This study involved the acquisition of EEG data from two distinct participant groups: individuals with normal vision (control group) and those diagnosed with amblyopia. Data were gathered under four specific visual conditions: EC, EO, DEC, and NDEC. This approach enables a comparison of neural activity, particularly alpha rhythms, between the two groups across varying states of visual conditions.
2. *Data Analysis*: The collected EEG data were subjected to analysis employing both traditional signal processing techniques and ML methodologies. FFT was utilized to extract frequency-domain features, with a focus on alpha rhythms (8-12 Hz). Additionally, ML classifiers were implemented to classify participants based on their EEG features. Feature importance analysis and permutation tests were conducted to evaluate the robustness and significance of the identified predictors, ensuring the reliability of the classification results.

2.2 Objectives

1. *Identify Differences*: The primary objective is to identify and characterise differences in alpha rhythms between individuals with normal vision and those with amblyopia. This includes examining variations in alpha wave frequency, and power, as well as other relevant frequency-domain features derived from the data analysis.
2. *Understand Neural Mechanisms*: By analysing the identified differences in neural oscillatory activity, the study aims to gain deeper insights into the neural mechanisms underlying amblyopia. Specifically, the research seeks to understand how amblyopia affects neural activity within the V1 and its contribution to the generation and modulation of alpha rhythms.
3. *Develop Diagnostic and Therapeutic Approaches*: Leveraging the findings from EEG feature analysis and ML classification, the study aims to develop non-invasive diagnostic biomarkers for amblyopia. The identification of specific EEG signatures associated with amblyopia could facilitate early diagnosis and inform the creation of targeted therapeutic interventions, thereby enhancing clinical outcomes for affected individuals.

Motivation Behind the Research:

1. *Advance Scientific Understanding*: Amblyopia affects approximately 1-5% of the population, yet the neural mechanisms underlying this condition remain insufficiently understood. This research endeavours to bridge this knowledge gap by investigating the impact of amblyopia on EEG alpha rhythms, thereby contributing to the broader scientific comprehension of this common visual disorder.
2. *Improve Patient Outcomes*: By identifying reliable EEG biomarkers for amblyopia, the study has the potential to revolutionise diagnostic practices, enabling earlier and more accurate detection of the condition. Furthermore, the insights gained from understanding the neural alterations in amblyopia could inform the development of innovative therapeutic approaches, ultimately improving visual and cognitive outcomes for patients.

3 Methods

3.1 Participants

A total of 13 participants were recruited for this study, including 9 males and 4 females, with ages ranging from 21 to 65 years. Our cohort included 6 participants with normal or corrected-to-normal vision and 7 individuals diagnosed with amblyopia. Prior to their participation, everyone provided written, informed consent as per the ethical guidelines approved by the University of Auckland Human Participants Ethics Committee (protocol #021040).

To determine ocular dominance, we administered the Pelli-Robson test and Contrast Sensitivity [47], using a letter size of 20/125, corresponding to 0.8 LogMAR (approximately 1 degree of visual angle). This test identified 5 participants as right-eye dominant and 1 as left-eye dominant among those with normal or corrected-to-normal vision. Among the amblyopic patients, 3 had their fellow eye on the left side and 4 on the right side. Additionally, all participants underwent a standardized Early Treatment Diabetic Retinopathy Study (ETDRS) test to measure visual acuity [50].

The results indicated that participants with normal or corrected-to-normal vision, as well as the fellow eyes of amblyopic patients, fell within the normal range of visual acuity, defined as between -0.1 and 0.2 LogMAR. In contrast, the amblyopic eyes exhibited visual acuity levels exceeding 0.3 LogMAR. Grating acuity was derived using a characteristic frequency of 2.5 cycles per letter. These findings establish a robust baseline for distinguishing between normal visual function and the impaired visual acuity associated with amblyopia, facilitating a more precise comparison and interpretation of EEG data across different visual conditions.

3.2 Experimental Design

The EEG data was collected using multi-channel recordings from both amblyopic patients and control subjects. The recordings were stored in Binary Data Format (BDF) files, each containing 40 seconds of EEG data, structured into four distinct 10-second segments:

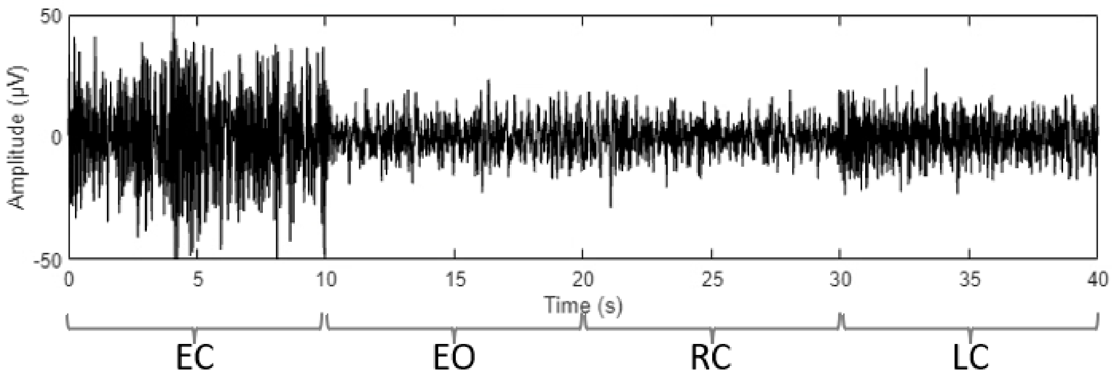

Figure 9 Example of a 40s recording from patient A9, divided by the distinct 10-second segments. EC: both eyes closed, EO: both eyes open, RC: right eye closed and LC: left eye closed.

1. *Both Eyes Closed (EC)*: The first 10 seconds recorded EEG activity with the participant's eyes fully closed.
2. *Both Eyes Open (EO)*: The next 10 seconds captured the EEG data with the participant's eyes open.
3. *Right Eye Closed (RC)*: In the final 10 seconds, the participant's right eye was closed while the left eye remained open.
4. *Left Eye Closed (LC)*: During the third 10 seconds, the participant's left eye was closed while the right eye remained open.

Given that we are working with amblyopic patients, where the amblyopic eye can be either the right or left eye, it is crucial to determine which eye is the fellow and the amblyopic. This process involved assigning specific labels to the LC and RC segments, reflecting the outcomes of the eye dominance tests. This distinction is made using the clinical characteristics of the amblyopic cohort, particularly the delta Visual Acuity (VA) score, defined by the differences in lines.

3.3 EEG Recordings

Spontaneous brain activity was recorded using the ActiveTwo biopotential acquisition system (ActiveTwo MK2; BioSemi, Amsterdam, Netherlands). This system allows for high-fidelity EEG recordings, essential for capturing subtle neural signals. We adhered to the internationally recognized 10-20 system for electrode placement [48], ensuring that electrodes were positioned at standardized locations on the scalp, which is vital for comparing results across studies and participants.

The data was sampled at a high frequency of 2048 Hz, which provides a detailed temporal resolution necessary for analysing fast-changing neural dynamics. Afterwards, the signals were processed to enhance the quality of the data:

- Bandpass Filtering: The EEG signals were filtered to retain frequencies between 2 and 80 Hz. This range includes most of the relevant brainwave activity while excluding very low-frequency drifts and high-frequency noise.
- Notch Filtering: A notch filter was applied to remove power line noise (48-52 Hz), which is a common artifact in EEG recordings [49], ensuring cleaner data for analysis.

For our analysis, we focused on ‘O1’, ‘O2’ and 'Oz' channels, each located at the occipital lobe, a region critically involved in visual processing. The occipital lobe is also a critical region because it is the primary source for the observation of alpha waves. For instance, the choice of the 'Oz' channel is particularly relevant for studying visual perception and the effects of amblyopia, as this area is directly engaged in processing visual stimuli.

To further refine the data, we applied a third-order Butterworth filter to bandpass the signals between 1 Hz and 40 Hz. This filtering step was chosen to include the alpha band (8-12 Hz), which is known to be significant in visual processing. Notably, alpha waves are observed when the eyes are closed, with no visual stimuli present, in both amblyopic [38] and control [15] patients. This characteristic alteration in individuals underscores the importance of the alpha band in our analysis. The Butterworth filter is advantageous due to its maximally flat frequency response in the passband, providing minimal distortion to the EEG signal.

By employing these recording and processing techniques, we ensured that the EEG data was of high quality, facilitating accurate and reliable analysis of neural activity under different visual conditions.

3.4 Data Analysis: Time-Frequency Domain

The time-frequency analysis allows us to pinpoint moments when specific frequency bands, like the alpha, are most active or disrupted.

3.4.1 Spectrogram

To investigate the temporal and spectral characteristics of the EEG signals under different visual conditions, a time-frequency analysis was conducted using spectrograms.

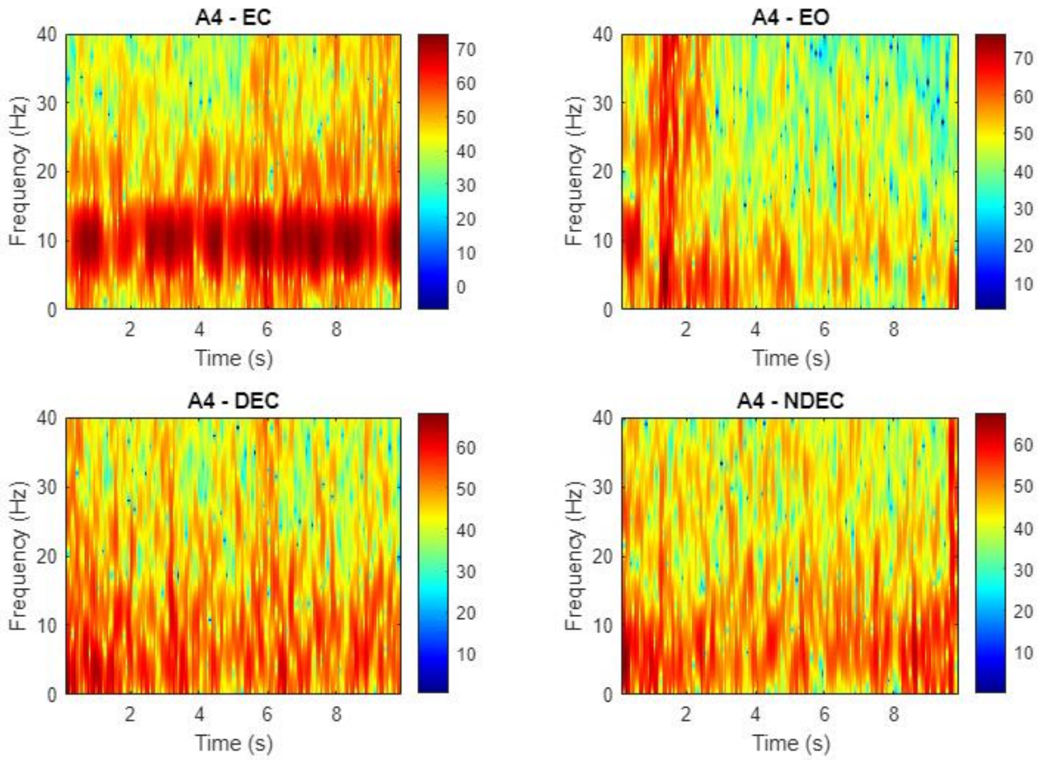

Figure 10 Time-frequency analysis plot from patient A4, divided by the EO, EC, DEC and NDEC sections.

The spectrograms were computed using the STFT with the following parameters to balance frequency resolution and temporal smoothness:

- Window Size: A Hamming window of length 512 samples was used to segment the EEG signals. This window size was chosen to provide sufficient frequency resolution while maintaining acceptable time resolution.
- Overlap: An overlap of 75% between consecutive windows was applied. This overlap enhances the temporal continuity of the spectrogram and reduces spectral leakage. (Overlap = 75% × 512 = 384 samples).
- Number of FFT Points: The number of points for the FFT was set to 16,384. This high value increases the frequency resolution of the spectrogram.
- Sampling Frequency: The sampling frequency (2048Hz) was obtained from each participant's data header to ensure accurate frequency scaling.
- Frequency Range of Interest: The analysis focused on frequencies from 0 to 40 Hz, encompassing key EEG bands such as delta, theta, alpha, beta, and lower gamma frequencies.

The power spectral density was converted to a decibel (dB) scale to enhance the visibility of changes in power across frequencies:

$$Power\ (dB) = 20 \log_{10}(|S(f,t)|) \qquad (I)$$

where S(f, t) is the spectrogram matrix.

3.5 Data Analysis: Time Domain

The time domain analysis offers a direct view of the brain's electrical activity as it unfolds over time. By examining the EEG signals, we can identify immediate patterns and anomalies, such as alterations in amplitude or latency that may distinguish amblyopic patients from controls.

3.5.1 Statistical Measures

This analysis focused on computing basic statistical measures for each condition and differences of statistical measures between the EO condition and the other conditions.

The first 2 seconds of each segment were discarded to eliminate potential artifacts or transient neural responses associated with the onset of the recording or condition change.

For each participant and condition, the following statistical measures were calculated:

- **Mean (μ):** The average value of the EEG signal.

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i \qquad (II)$$

- **Variance (σ^2):** The measure of signal dispersion.

$$\sigma^2 = \frac{1}{N-1}\sum_{i=1}^{N} (x_i - \mu)^2 \qquad (III)$$

- **Standard Deviation (σ):** The square root of variance.

$$\sigma = \sqrt{\sigma^2} \qquad (IV)$$

- **Skewness:** The measure of asymmetry in the signal distribution.

$$Skewness = \frac{\frac{1}{N}\sum_{i=1}^{N} (x_i - \mu)^3}{\sigma^3} \qquad (V)$$

- **Kurtosis:** The measure of the "tailedness" of the signal distribution.

$$Kurtosis = \frac{\frac{1}{N}\sum_{i=1}^{N} (x_i - \mu)^4}{\sigma^4} \qquad (VI)$$

To assess the changes between conditions, difference metrics were calculated by subtracting the statistical measures of the EO condition from those of the other conditions (EC, DEC, NDEC):

- **Difference in Mean (e.g., using EC):**

$$\Delta\mu_{EO-EC} = \mu_{EO} - \mu_{EC} \qquad (VII)$$

- **Difference in Standard Deviation (e.g., using DEC):**

$$\Delta\sigma_{EO-DEC} = \sigma_{EO} - \sigma_{DEC} \qquad (VIII)$$

- **Difference in Variance, Skewness, Kurtosis:** Computed similarly.

To evaluate whether there are statistically significant differences in EEG-derived measures between various visual conditions, we specifically compared the following pairs: EO-EC, EO-

DEC, and EO-NDEC. Each of these pairwise comparisons was conducted to identify any notable differences between the conditions.

Prior to selecting the appropriate statistical test, the normality of each dataset was assessed using the Kolmogorov-Smirnov test. If both datasets in a comparison were normally distributed ($p > 0.05$), a two-sample t-test or Mann-Whitney U test was applied.

3.5.2 Relevant Features

In this section, we aimed to classify different EEG conditions by applying ML techniques to time-series data recorded from the three EEG electrodes (O1, Oz, and O2). We utilised "tsfresh" [51] library to automatically extract a comprehensive set of features, including statistical measures, also frequency-domain characteristics. Here, we also removed the first 2 seconds from all conditions.

The dataset was split into training and testing sets using stratified sampling to preserve class distribution, with 75% allocated for training and 25% for testing. We applied feature selection using the ANOVA F-test to identify the top 50 most significant features, reducing dimensionality and focusing on the most informative attributes. This selection enhanced model performance by eliminating redundant or irrelevant features.

We evaluated several ML classifiers, including Random Forest (RF), Gradient Boosting (GB), Logistic Regression (LR), Support Vector Machine (SVM), Neural Network (NN), and XGBoost (XGB). For each classifier, we performed hyperparameter tuning using grid search combined with five-fold cross-validation to identify the optimal parameters. The models were trained on the selected features, and their performance was assessed on the test set using metrics such as accuracy, precision, recall, and F1-score. For models providing feature importance scores, we analysed the most influential features contributing to the classification.

To evaluate the statistical significance of the classification model's performance in distinguishing between control and amblyopic patients, we conducted permutation tests. Initially, we balanced the dataset by removing one amblyopic participant, resulting in equal numbers of control and amblyopic subjects.

We randomly permuted the labels across the entire dataset to disrupt any true associations between the EEG features and the labels. Since participant IDs encode label information (starting with 'A' for amblyopic and 'C' for control), we shuffled the IDs among existing participants. This was achieved by generating a random permutation of all unique participant IDs and mapping each original ID to a new one from this shuffled list. The EEG time-series data associated with each participant remained intact; only the labels were reassigned.

3.6 Data Analysis: Frequency Domain

The frequency domain analysis allows us to decompose the EEG signal into its constituent frequencies, providing a more detailed perspective on how different neural oscillations contribute to overall brain activity. In this phase of the analysis, our main focus is the alpha frequency band that are known to be associated with various cognitive and sensory processes.

This frequency-based exploration offers a different lens through which to view the data. While time domain analysis provides a snapshot of neural activity, frequency domain analysis uncovers the rhythmic patterns that govern brain function. It helps us understand whether the neural circuits in amblyopia are fundamentally altered in how they oscillate and synchronize during visual tasks. The first 2 seconds of each analysis is discarded to remove unwanted artifacts from previous conditions.

FFT was used to convert time-domain signals into frequency-domain spectra. It returns the amplitude and phase. By taking the absolute value we obtain the magnitude of each frequency component. Magnitudes of the spectra are converted to the logarithmic dB scale. When dealing with quantities like voltage, which are related to power by a square relationship, the conversion formula is:

$$Magnitude\ in\ dB = 20 * log_{10}\left(\frac{V_2}{V_1}\right) \tag{IX}$$

In EEG data analysis using the FFT, each frequency component's magnitude from FFT results is treated as $V2$, with $V1$ set to 1 µV as a reference, simplifying dB conversion. This conversion evaluates each component relative to a normalisation reference and presents the data on a logarithmic scale. This scale is intuitive for visualizing and analysing subtle power differences across frequencies. Additionally, it compresses the value range, making the data easier to handle and crucial for effectively detecting patterns or abnormalities in EEG readings.

3.6.1 Reactivity

Building upon Wieneke's work [15] on normative spectral data of the alpha rhythm in male adults, our analysis focused on the computation of reactivity to quantify the difference in EEG spectral power density at the peak alpha frequency between distinct conditions, such as EO and EC. Wieneke's findings underscored the significance of the alpha band (8–12 Hz) in EEG studies, which motivated us to emphasize in this frequency range.

We extracted the power spectral density (PSD) for the visual conditions, including EC, EO, DEC, and NDEC. Alongside this, we retrieved the frequency axis for each set of EEG data. Our primary focus was on identifying the frequency indices that fell within the alpha band, specifically between 8–12 Hz.

To establish the peak frequency, we calculate the power spectrum for the EC condition and get the peak within the alpha band (8–12 Hz). Recognising that it exists slight variations in peak alpha frequency between conditions, we defined an acceptable frequency range of ±0.5 Hz around this peak. This allowed us to compare the EO condition at closely corresponding frequencies. We focused on EO as the baseline for analysis, following Wieneke's methodology. Figure 11 explains why choosing exactly the same frequency wouldn't capture the best peaks for EO. This is due to slight shifts in peak frequencies between conditions.

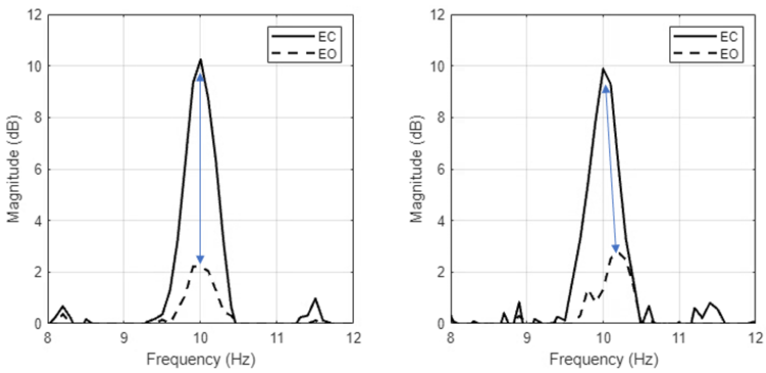

Figure 11 Two side-by-side plots using illustrative data to compare EC and EO power spectra. Left plot with aligned peaks at 10 Hz, right plot with EO peak misaligned at 10.2 Hz.

We computed reactivity as the peak difference (EO minus EC) in dB, measuring how much the alpha power decreases upon eye opening. We applied smoothing (moving average of 5) to the spectral data before plotting to reduce noise and clarify trends.

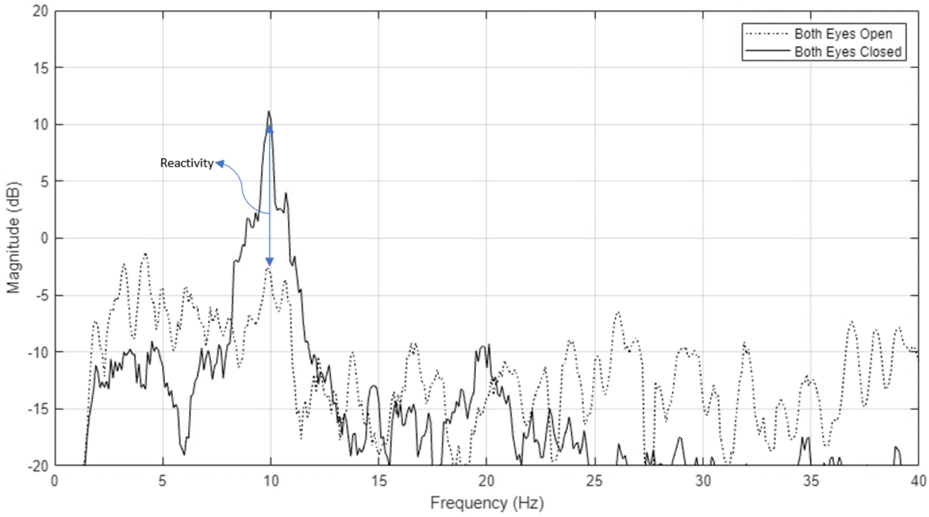

Figure 12 Example of reactivity (EO to EC) measured from participant A4.

Figure 12 shows an example of reactivity from EO to EC for participant A4. The negative reactivity value indicates higher power during EC compared to EO, reflecting the typical increase in alpha power when eyes are closed.

We assessed the statistical significance of differences between the groups using two-sample t-tests with equal variances, testing the hypothesis of no difference in mean reactivity between pairs of experimental conditions. We computed p-values for comparisons between EO-EC and EO-DEC, EO-EC and EO-NDEC, and EO-DEC and EO-NDEC.

3.6.2 Bandwidth

For the bandwidth we also used the resulting spectra, which is converted to decibels (dB) using the FFT and smoothed using a moving average filter to reduce noise.

Figure 13 demonstrates the spectral characteristic of the EEG signal from EC condition for subject A9.

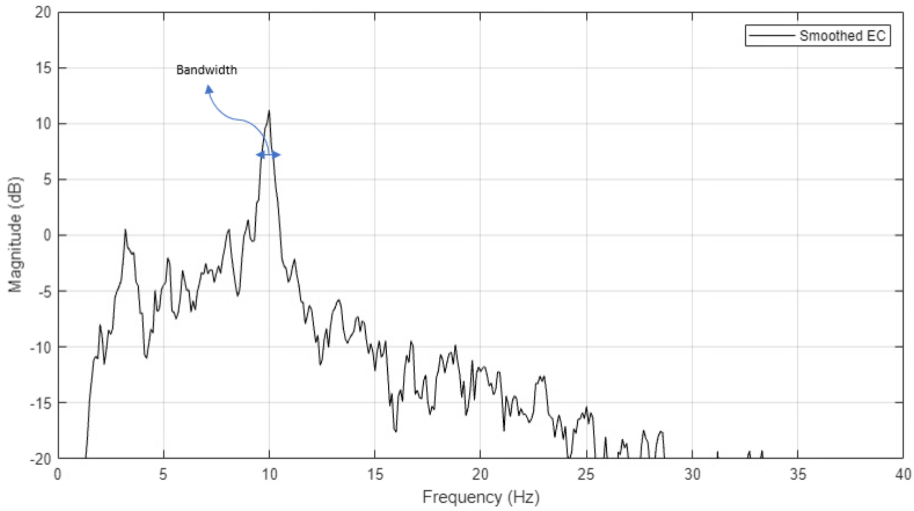

Figure 13 Example of bandwidth from EC measured from participant A9.

To quantify the spectral bandwidths of EEG signals across different conditions, we employ the following steps:

1. Identify the Frequency Range: For each condition, the frequency axis is constrained to the 8-12 Hz range, which corresponds to the alpha band typically associated with relaxation and cognitive activities.

2. Peak Detection: Within this range, the peak power (in dB) and its corresponding frequency are identified. This peak represents the most dominant oscillation within the alpha band for the given condition.

3. Calculate -3 dB Bandwidth: The bandwidth is defined as the frequency range where the power spectrum falls by 3 dB from its peak value. This is determined by finding the points on either side of the peak where the spectrum first crosses the target level of (peak dB - 3).

4. Determine Bandwidth Limits: The frequencies at these crossing points are recorded as the lower and upper bounds of the bandwidth. The bandwidth is then calculated as the difference between these bounds.

5. Handle Edge Cases: If the spectrum does not cross the -3 dB point within the 8-12 Hz range, the bandwidth is marked as undefined or not a number (NaN), indicating a lack of clear peak or insufficient signal strength in this range.

3.6.3 Skewness

We initiated the analysis by defining the frequency range of interest, focusing on two primary intervals: 0-40 Hz, which encompasses the full range of typical EEG activity, and 8-12 Hz, targeting the alpha band.

For each dataset, the frequency axis and spectral data were extracted for the four conditions (EC, DEC, NDEC, and EO). We compared the skewness of EEG spectral data between different experimental conditions by treating the EO condition as the baseline. This approach facilitates the assessment of how these conditions differ in terms of spectral asymmetry.

For each condition, the differences in skewness between EO and EC, EO and DEC, and EO and NDEC are calculated for both frequency ranges. Statistical testing is then applied to determine whether significant differences exist between conditions. Paired t-tests are conducted within each frequency range (0-40 Hz and 8-12 Hz) to compare the differences between EO and EC, EO and DEC, and EO and NDEC.

3.6.4 Kurtosis

We followed the same approach as described in the previous section (skewness). Calculating kurtosis for two frequency ranges, 0-40 Hz and 8-12 Hz. This analysis aimed to identify whether certain visual conditions elicited changes in spectral distribution properties.

We also calculated the difference in kurtosis between EO and the other three conditions (EC, DEC, NDEC) for both frequency ranges. These differences are computed separately for the amblyopia and control groups, allowing for a detailed comparison between these groups.

Statistical tests (e.g., independent t-tests, Mann-Whitney U tests) were conducted to assess whether observed differences were statistically significant.

3.6.5 Multivariate Pattern Analysis

To eliminate transient artifacts the first 2 seconds of each recording were also discarded. To investigate the impact of different epoch lengths on classification performance, the EEG data were segmented into non-overlapping epochs of varying durations:

- One non-overlapping 8-second epoch, resulting in a total of 3 epochs per condition per participant (1 epoch x 3 electrodes).
- Two non-overlapping 4-second epochs, resulting in a total of 6 epochs per condition per participant (2 epochs × 3 electrodes).
- Four non-overlapping 2-second epochs, resulting in a total of 12 epochs per condition per participant (4 epochs × 3 electrodes).

For each epoch, spectral analysis was performed. The PSD was calculated, and the mean power within the following frequency bands was extracted:

- Theta Band (4–8 Hz): Often linked to drowsiness, memory, and navigation processes.
- Alpha Band (8–12 Hz): Associated with relaxed wakefulness and visual processing.
- Beta Band (16–30 Hz): Related to active thinking, concentration, and motor activity.
- High-Gamma Band (36–40 Hz): Linked cognitive processing and higher-order visual functions.

The power values were converted to dB to normalise the data.

Each epoch was labelled with:

- Subject Group: Amblyopia patient (1) or control subject (0).
- Electrode Number: O1 (15), Oz (16), or O2 (17).
- Subject Identifier: A unique ID for each participant.
- Condition: One of the four visual conditions.

This labelling allowed for group-based analyses and ensured that data could be aggregated or separated based on electrodes and conditions as needed.

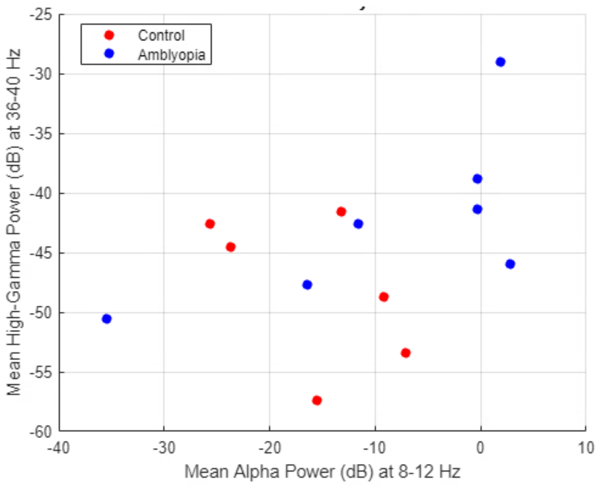

Figure 14 Mean Alpha vs High-Gamma Power for EC (First 2s removed) from Oz electrode. One non-overlapping 8-second epoch.

We also employed mean alpha power and bandwidth to classify amblyopia patients and control subjects. Bandwidth calculated as the difference between the upper and lower -3 dB cutoff frequencies.

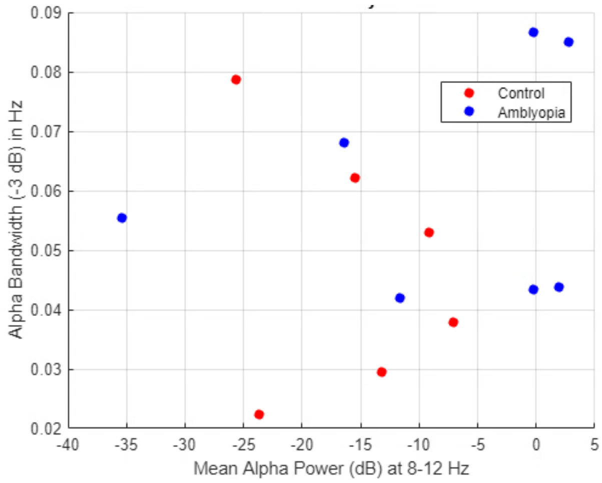

Figure 15 Mean Alpha Power vs Bandwidth for EC (First 2s removed) from Oz electrode. One non-overlapping 8-second epoch.

A Random Forest classifier was employed to distinguish between amblyopia patients and control subjects based on the extracted features. The classifier was configured with 100 decision trees. Leave-one-out cross-validation (LOO CV) strategy was used.

3.6.6 Feature Matching

The 8 seconds EEG data were segmented into four non-overlapping 2-second epochs, resulting in a total of 16 epochs per participant per electrode (4 conditions × 4 epochs).

Each EEG signal was subjected to z-score normalisation to standardise the data across participants and electrodes. This normalisation was performed using the formula:

$$Z - score = \frac{X - \mu}{\sigma} \quad \text{(X)}$$

where X is the raw EEG signal, μ is the mean, and σ is the standard deviation of the signal.

The FFT was computed for each windowed segment to obtain the frequency spectrum. The ratio of alpha band power to reference band power was calculated for each epoch and converted to dB using the formula:

$$Amplitude\ Ratio\ (dB) = 20log_{10}\left(\frac{Power_{alpha}}{Power_{reference}}\right) \quad (XI)$$

where alpha band power was calculated by summing the amplitudes of the FFT components within the 8–12 Hz range. This higher frequency band (36-40Hz) was selected as a reference to normalise the alpha power, accounting for individual variations in overall EEG amplitude.

The extracted amplitude ratios were organised into a feature matrix, where each row represented an epoch from a participant, and each column corresponded to a specific condition and electrode combination. Given that each participant contributed 4 epochs per condition (after discarding the first 2 seconds), the total number of samples in the feature matrix was 13 participants×4 epochs=52 samples per electrode, and 52×3 electrodes=156 samples overall. For the classification task, a subset of features was selected based on their potential discriminative power.

A Random Forest classifier was employed to perform the binary classification between amblyopia patients and control subjects. A 13-Fold LOO CV was implemented to assess the model.

A confusion matrix was constructed by comparing the predicted labels with the true labels, categorizing outcomes as True Positives (TP), True Negatives (TN), False Positives (FP), or False Negatives (FN).

The classifier's performance was evaluated using accuracy and MCC described by the formula:

$$MCC = \frac{(TPxTN) - (FPxFN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \qquad (XII)$$

In addition, to assess the statistical significance of the classifier's performance, permutation (shuffle) tests were conducted.

4 Results

Now we focus on the assessment of alpha rhythms in subjects with and without amblyopia by analysing the empirical findings derived from the EEG data analysis. The analyses were conducted following the methods outlined in the previous chapter, leveraging time-frequency domain, time domain and frequency domain analyses to explain the characteristics of EEG signals pertinent to amblyopia.

4.1 Data Analysis: Time-Frequency Domain

4.1.1 Spectrogram

From Figure 10 and 16 we notice that there is a distinct concentration of power in the alpha frequency range (8-12 Hz), especially in the EC condition. This observation aligns well with known characteristics of the alpha band, which is generally dominant when the eyes are closed, and the subject is relaxed, and brain activity is less influenced by visual stimuli.

Figure 16 presents spectrograms for multiple participants and in sequence the EC and EO conditions.

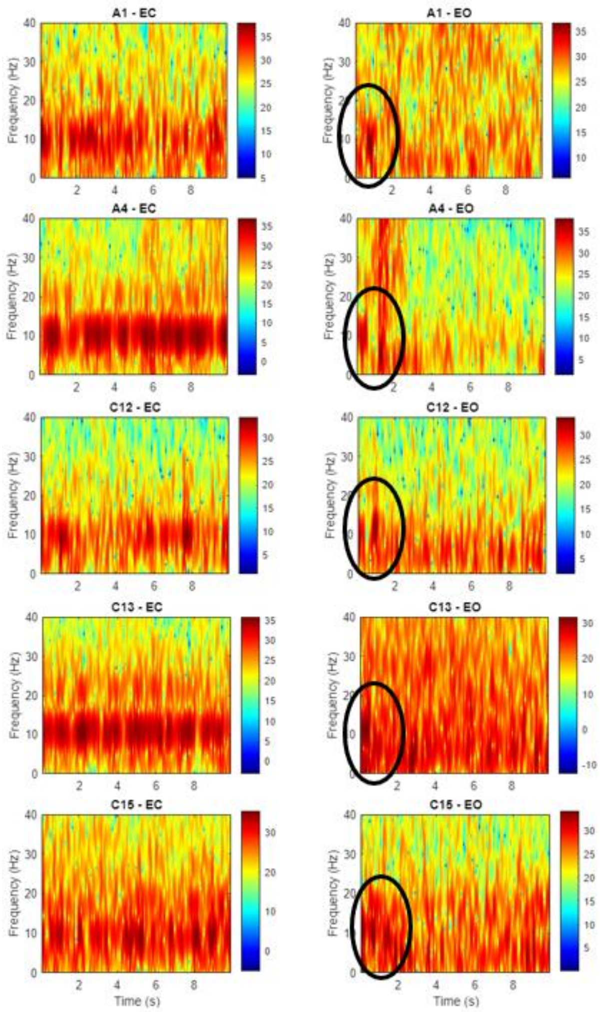

Figure 16 Spectrogram for patients A1, A4, C12, C13 and C15, from section EC and EO. Black circles on sections EO, highlights 8-12Hz artifacts remaining from EC previous condition.

It's observed a carryover of activity in the alpha band during the initial 1-2 seconds of the EO condition. This is very likely an artifact due to a transition effect as the participant shifts from having their eyes closed to eyes open. This transitional persistence is common, as it takes a moment for the brain to adjust and suppress alpha activity when visual processing begins. The necessity to remove the first 2 seconds in other analyses becomes evident, as this helps avoid potential misinterpretations that would result from alpha activity or other characteristics leaking into other segments.

Worth mentioning that the power distribution across higher frequency bands (20-40 Hz) is also visible across all conditions, which might be attributed to ongoing cognitive processing. The power at higher frequencies (beta and gamma) seems relatively uniform across these conditions, which may indicate sustained alertness or attention.

4.2 Data Analysis: Time Domain

4.2.1 Statistical Measures

The pairwise statistical analysis results provided allow us to assess whether there are significant differences between different pairs of EEG conditions (EO-EC, EO-DEC, and EO-NDEC) based on various statistical metrics.

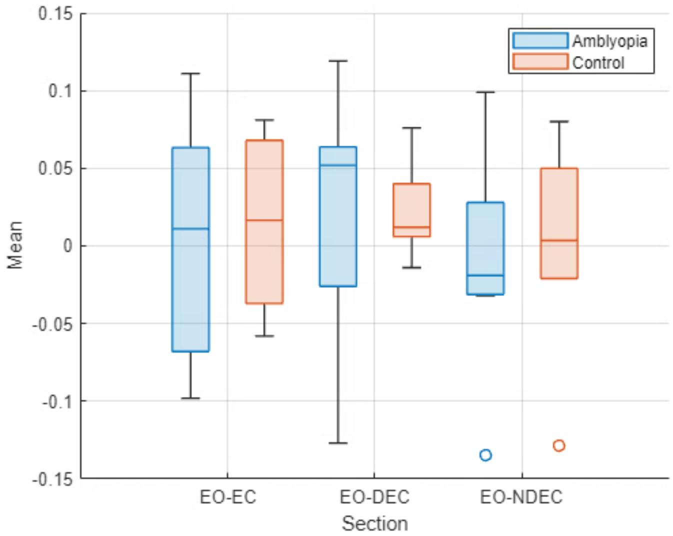

Figure 17 Boxplot with Amblyopia and Control mean with sections containing the difference between EO and EC, DEC and NDEC.

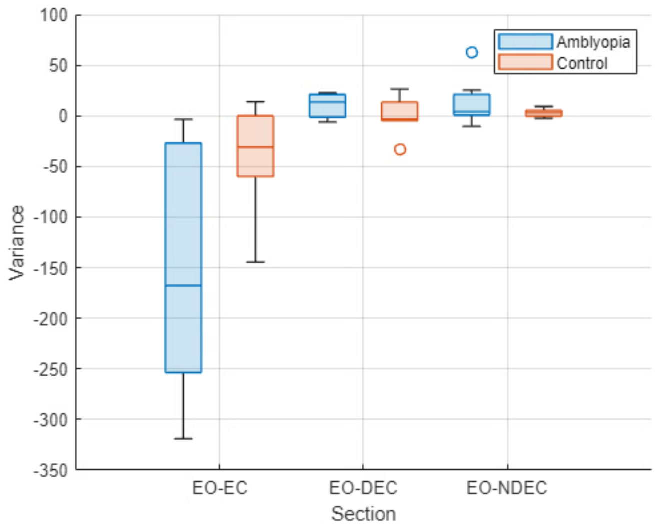

Figure 18 Boxplot with Amblyopia and Control variance with sections containing the difference between EO and EC, DEC and NDEC.

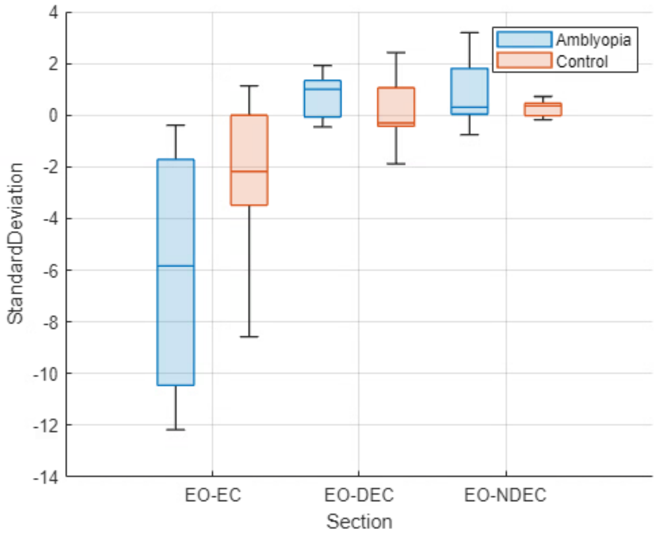

Figure 19 Boxplot with Amblyopia and Control standard deviation with sections containing the difference between EO and EC, DEC and NDEC.

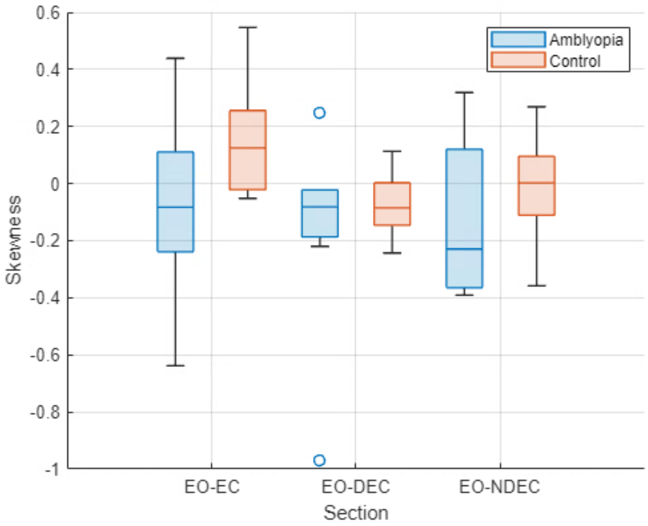

Figure 20 Boxplot with Amblyopia and Control skewness with sections containing the difference between EO and EC, DEC and NDEC.

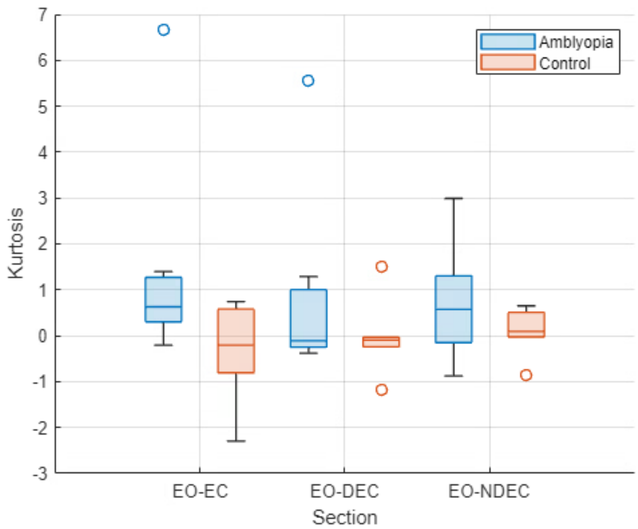

Figure 21 Boxplot with Amblyopia and Control kurtosis with sections containing the difference between EO and EC, DEC and NDEC.

- Mean, Skewness and Kurtosis: No significant differences were found between any pair of conditions.
- Variance and Standard Deviation: Significant differences were observed between EO-EC vs EO-DEC and EO-EC vs EO-NDEC, but not between EO-DEC and EO-NDEC.

Table 1 p-values of statistical measures

p-values	Mean	Variance	Standard Deviation	Skewness	Kurtosis
EO-EC vs EO-DEC	0.86	**0.001**	**0.001**	0.14	0.84
EO-EC vs EO-NDEC	0.61	**0.005**	**0.0005**	0.36	0.79
EO-DEC vs EO-NDEC	0.20	0.72	0.74	0.92	0.98

The results suggest that while the central tendency (mean) and the shape of the distributions (skewness and kurtosis) do not differ significantly across the conditions, the variability (variance and standard deviation) of EEG signals is notably different when comparing EO-EC with EO-DEC and EO-NDEC, indicating potential differences in the dispersion of neural activity across these states.

Table 2 p-values for variance and standard deviation, compared between amblyopia and control

p-values	Variance	Standard Deviation
EO-EC (Amblyopia vs Control)	0.14	0.23
EO-DEC (Amblyopia vs Control)	0.36	0.36
EO-NDEC (Amblyopia vs Control)	0.53	0.36

The group statistical analysis comparing amblyopia and control groups for variance and standard deviation shows that there are no significant differences between these groups across the various conditions (EO-EC, EO-DEC, EO-NDEC). The p-values for both variance and standard deviation comparisons are not above the significance threshold (0.05), indicating that the variability and spread of EEG signals are similar between the groups in each condition.

4.2.2 Relevant Features

The results highlighted the efficacy of ensemble methods like RF, NN and XGB in handling complex EEG time-series data. The analysis also provided insights into the most critical features influencing classification, offering potential implications for automated EEG analysis.

Table 3 Performance metrics of ML classifiers on EEG data

Classifier	Accuracy	Precision	Recall	F1-score	Important Features
RF	0.77	0.77	0.77	0.77	O1__fft_coefficient, Oz__fft_coefficient, O2__fft_coefficient, O1__augmented_dickey_fuller, O1__ar_coefficient
GB	0.69	0.69	0.69	0.69	O1__fft_coefficient, Oz__fft_coefficient, O2__fft_coefficient, O1__ar_coefficient
XGB	**0.85**	**0.85**	**0.85**	**0.85**	O1__fft_coefficient, Oz__fft_coefficient, O2__fft_coefficient, O1__augmented_dickey_fuller, O1__ar_coefficient
LR	0.77	0.78	0.77	0.77	N/A
NN	0.77	0.78	0.77	0.77	N/A
SVM	0.54	0.54	0.54	0.54	N/A

Logistic regression, NN and SVM does not provide feature importances directly.

The “fft_coefficient” captures frequency-domain information, such as power at specific frequencies. FFT coefficients from electrodes O1, Oz, and O2 were consistently important across tree-based models.

The “ar_coefficient” represents the coefficients of an autoregressive (AR) model fitted to the EEG signal. It captures temporal dependencies and patterns. Another occasional feature is the “augmented_dickey_fuller”, which is a statistical test used to check for stationarity in a time series. The augmented Dickey-Fuller statistic and autoregressive coefficients from the O1 electrode suggest that temporal dynamics and signal stationarity can contribute to the classification.

XGB achieved the highest accuracy of 85% among all classifiers, indicating its effectiveness in handling the complexities of EEG data. The consistent values across precision, recall, and F1-score suggest a balanced performance in correctly identifying both amblyopia patients and control subjects.

The NN matched the performance of LR and RF with an accuracy of 77%. The best NN configuration involves a single hidden layer of 1000 neurons, the Tanh activation function, stochastic gradient descent optimizer, and a learning rate of 0.0001.

SVM exhibited the lowest performance, barely surpassing random chance. This may be due to limitations in handling non-linear patterns.

The importance of frequency-domain (fft_coefficient) and time-series features (augmented_dickey_fuller, ar_coefficient) suggests that specific neural signatures associated with amblyopia can be captured and utilised for classification.

The classification metrics obtained after the permutation test are presented in Table 4.

Table 4 Classification performance metrics after 1050 random permutations with 95% confidence-interval

Classifier	Accuracy	Precision	Recall	F1-score	MCC
RF	0.52 [0.5, 0.54]	0.55 [0.53, 0.58]	0.6 [0.57, 0.63]	0.57 [0.54, 0.59]	0.03 [-0.53, 0.56]
GB	0.51 [0.49, 0.53]	0.56 [0.53, 0.59]	0.53 [0.5, 0.56]	0.53 [0.5, 0.55]	0.02 [-0.55, 0.55]
XGB	0.51 [0.49, 0.53]	0.55 [0.52, 0.57]	0.57 [0.53, 0.6]	0.55 [0.52, 0.57]	0.01 [-0.56, 0.56]
LR	0.52 [0.49, 0.54]	0.56 [0.53, 0.58]	0.57 [0.54, 0.6]	0.55 [0.53, 0.57]	0.03 [-0.55, 0.54]
NN	0.5 [0.48, 0.52]	0.55 [0.52, 0.57]	0.54 [0.51, 0.57]	0.53 [0.51, 0.55]	0.0 [-0.53, 0.5]
SVM	0.53 [0.51, 0.55]	0.56 [0.54, 0.57]	0.62 [0.59, 0.65]	0.57 [0.55, 0.6]	0.04 [-0.52, 0.54]

As expected under the null hypothesis, all classifiers exhibited performance close to chance level with an accuracy of 0.50, which aligns with the expected accuracy for random guessing in a binary classification task.

The collective results demonstrate that, when the true relationships between the EEG features and the labels are disrupted, the classifiers are generally unable to distinguish between control and amblyopic participants. This outcome confirms that the original models' performances were not artifacts of data leakage or random chance. Therefore, the permutation test was successful in validating the statistical significance of our initial findings.

4.3 Data Analysis: Frequency Domain

4.3.1 Reactivity

The results of the statistical tests provide insights into the differences in EEG reactivity between the experimental conditions. The p-values obtained from the t-tests are as follows:

- P-value between EO-EC and EO-DEC: 0.00 - The zero p-value indicates a statistically difference in reactivity between the EO-EC and the EO-DEC reactivities.

- P-value between EO-EC and EO-NDEC: 0.00 - Like the first comparison, the p-value also indicates a statistically significant difference in EEG reactivity between the EO-EC and EO-NDEC.

- P-value between EO-DEC and EO-NDEC: 0.47 - This p-value is substantially higher and indicates that there is no statistically significant difference in EEG reactivity between the DEC and NDEC conditions.

Both EO-DEC and EO-NDEC conditions alter brain reactivity when compared to EO-EC. However, there is no significant difference between the DEC and NDEC conditions themselves, suggesting similar impacts on EEG reactivity from these experiments.

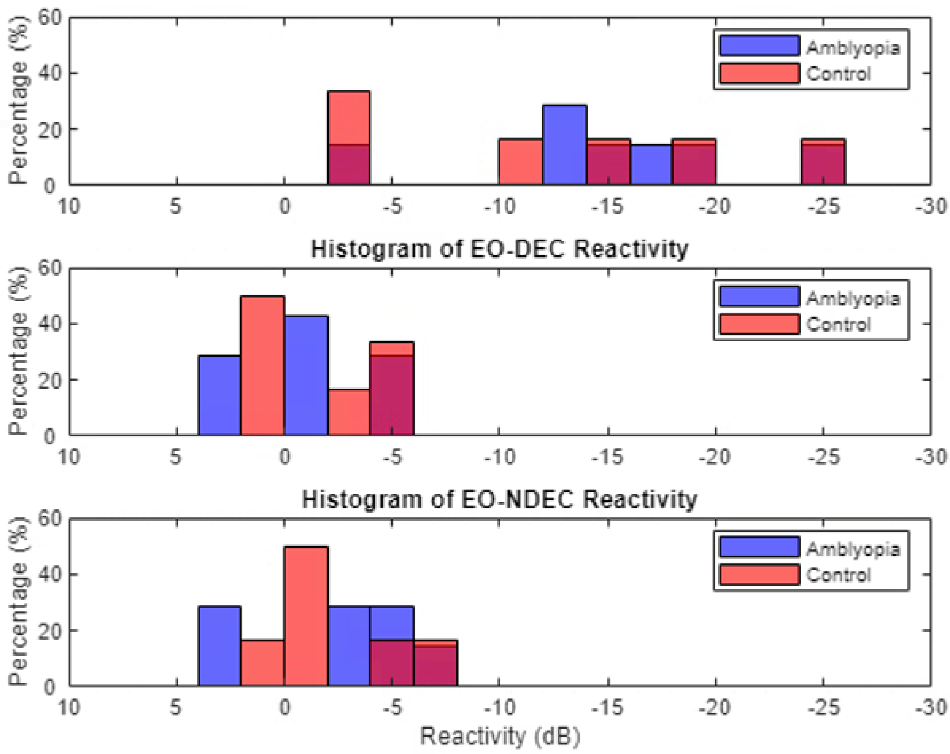

Figure 22 Histograms with Amblyopia and Control reactivities. 1st: difference between EO and EC; 2nd: difference between EO and DEC; 3rd: difference between EO and NDEC.

Figure 22 depict the distribution of reactivity values for different conditions among amblyopia and control groups.

For the EO-EC reactivities, both groups have a broad distribution, extending from about -2 dB to -25 dB, indicating great variability and suggesting a more pronounced decrease in activity when both eyes are closed.

For the EO-DEC and EO-NDEC reactivities, the histograms do not exhibit a clear decrease in reactivity for both the amblyopia and control groups, suggesting that significant reactivity is not universally present across these conditions. This indicates a potential uniformity in the response to dominant and non-dominant eye closure, where neither condition markedly alters the alpha rhythm, highlighting a possible baseline stability in neural processing irrespective of eye dominance.

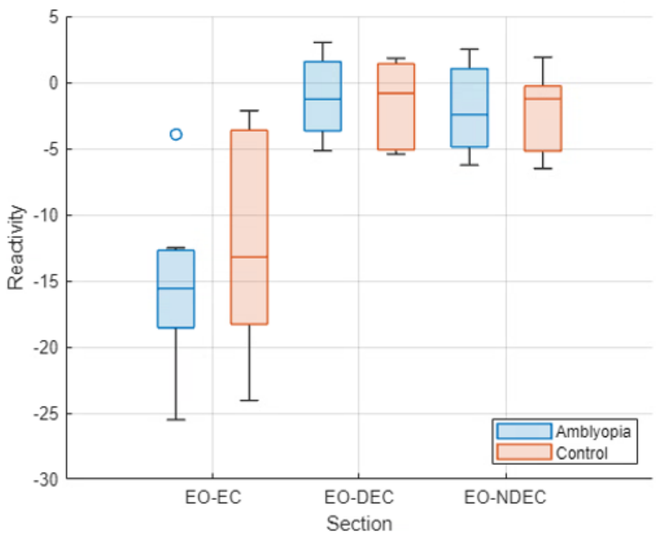

Figure 23 Boxplot with Amblyopia and Control reactivity with sections containing the difference between EO and EC, DEC and NDEC.

For each of the experimental reactivity, EO-EC, EO-DEC, and EO-NDEC, we computed the p-values using the two-sample t-test.

- EO-EC: The p-value of 0.52 suggests that there is no statistically significant difference in EEG reactivity. This indicates that the general condition of eye closure does not differentially affect the EEG reactivity between these two groups.

- EO-DEC: With a p-value of 0.83, the results also indicate no significant difference in EEG reactivity.

- EO-NDEC: The p-value of 0.94 further supports the lack of statistically significant differences in EEG reactivity between the groups when the non-dominant eye is closed.

This implies that the neurological responses measured through EEG under these specific conditions do not vary significantly based on the presence of amblyopia.

Based on Figure 24 from the reactivity data, no distinct relationship between EO-EC and EO-DEC reactivity measures can be observed. The data suggests that these two reactivity measures are not significantly correlated, indicating that individuals who exhibit high reactivity in the EO-EC condition do not necessarily exhibit high reactivity in the EO-DEC condition. This lack of correlation implies that the underlying neural mechanisms governing the difference from eyes open to eyes closed and from eyes open to dominant eye closed may not share common pathways or may be influenced by different factors.

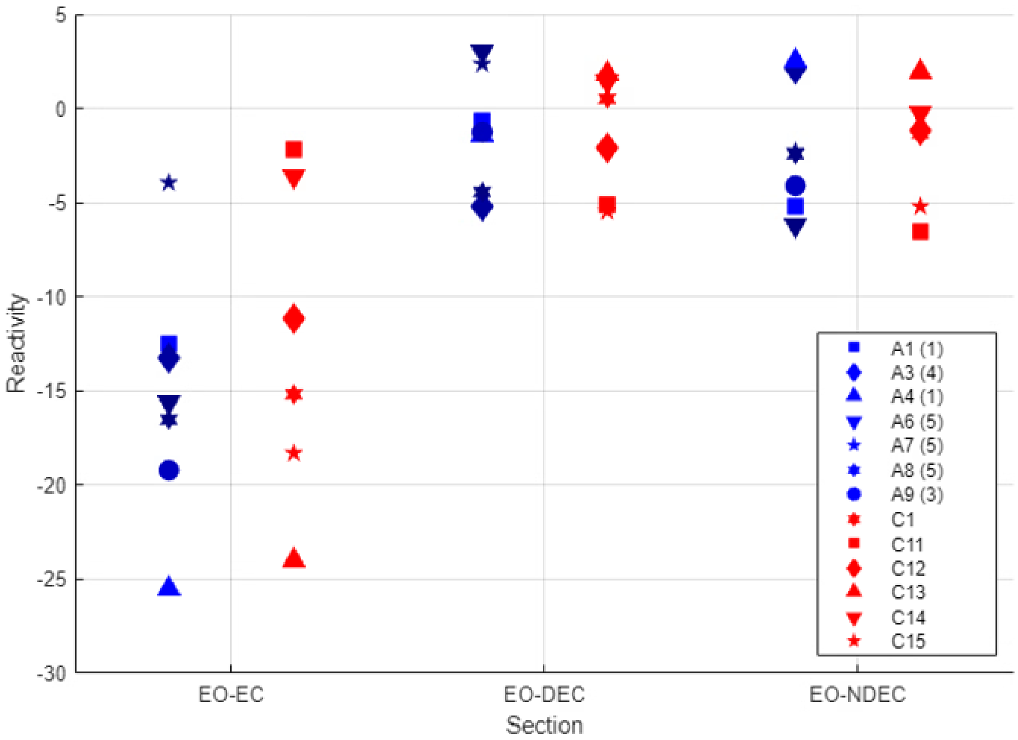

Figure 24 Scatter with Amblyopia and Control reactivities separated by sections and intensity levels based on visual acuity test (1-low, 5-high).

When examining the intensity levels derived from the amblyopia difference lines of the visual acuity test, no clear pattern emerges. The scatter plot does not show a systematic variation in reactivity associated with different intensity levels. This absence of a discernible pattern suggests that the degree of visual impairment, as quantified by the difference in visual acuity, does not directly influence the reactivity measures in EO-EC or EO-DEC conditions. This outcome might indicate that the mechanisms responsible for reactivity changes are more complex and are not solely dependent on the severity of amblyopia as measured by visual acuity differences.

4.3.2 Bandwidth

The bandwidths vary, with EC conditions showing narrow bandwidths (indicating focused and strong alpha activity) and others (EO, DEC, NDEC) displaying broader or undefined bandwidths (suggesting more dispersed or weaker alpha activity). Overall, the average bandwidth across all conditions and subjects is 0.99 Hz, with a standard deviation of 0.41 Hz.

In our experiment, no significant differences in bandwidths were observed between any of the conditions.

- EO-EC (p-value: 0.82): The comparison between eyes open and eyes closed conditions revealed a p-value of 0.82, which is well above the 0.05 significance threshold.

- EC-DEC (p-value: 0.93): Also, no statistically significant difference in bandwidths between these conditions.

- EC-NDEC (p-value: 0.58): Similarly, the comparison between bandwidths in the eyes closed and non-dominant eye closed conditions resulted in a p-value of 0.58, which is also above limit threashold.

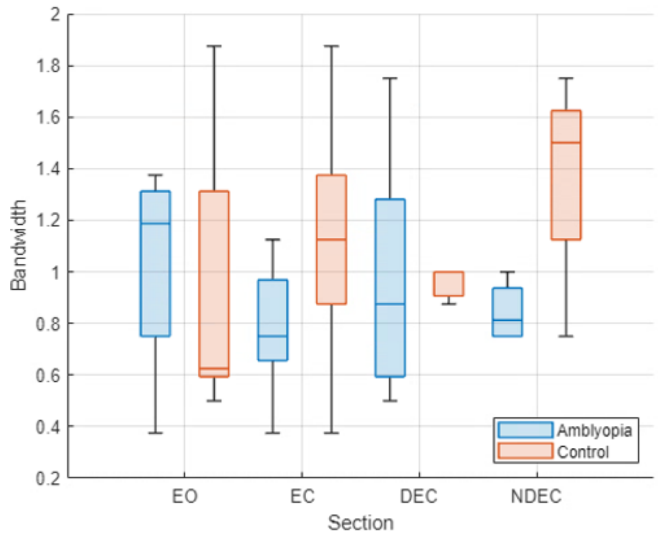

Figure 25 Boxplot with Amblyopia and Control bandwidth with sections containing EO, EC, DEC and NDEC.

Regarding the difference across groups, the NDEC bandwidth p-value (0.055) is just above the significance level. There may be underlying factors causing a difference in NDEC bandwidth between the groups that are not strong enough to be statistically confirmed with the current sample size.

However, for our experiments, there are no significant differences in the alpha band bandwidths between the amblyopic and control groups across all conditions (EO, EC, DEC, NDEC).

4.3.3 Skewness

Significant differences in skewness were observed when comparing EO-EC to EO-DEC, and EO-NDEC, across both the alpha band and the full spectrum.

Table 5 Statistical comparisons of skewness differences between condition pairs in the alpha band (8–12 Hz)

Comparison of Differences	p-value
EO-EC vs. EO-DEC	**0.0043**
EO-EC vs. EO-NDEC	**0.0001**
EO-DEC vs. EO-NDEC	0.71

Table 6 Statistical comparisons of skewness differences between condition pairs in the full spectrum (0–40 Hz)

Comparison of Differences	p-value
EO-EC vs. EO-DEC	**0.0001**
EO-EC vs. EO-NDEC	**0.0001**
EO-DEC vs. EO-NDEC	0.83

The significant differences involving the EO-EC comparisons highlight that closing both eyes alters the EEG skewness differently than closing a single eye. These results suggest that bilateral eye closure has a unique impact on neural activity patterns, as reflected by changes in skewness, which are not replicated by unilateral eye closure. The absence of significant differences between the dominant and non-dominant eye closures further indicates that the specific eye closed does not play a significant role in affecting skewness differences.

The skewness of the EEG signal was also analysed to assess asymmetries in the spectral power distribution within and between groups across different visual conditions.

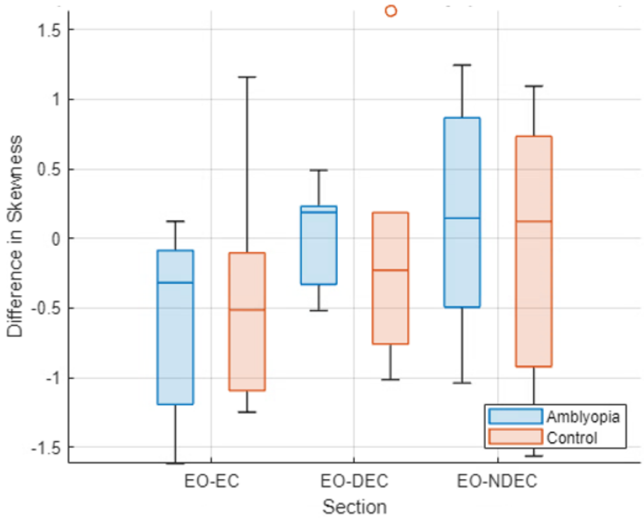

Figure 26 Boxplot with Amblyopia and Control skewness with sections containing the difference between EO and EC, DEC and NDEC in the alpha band (8-12Hz).

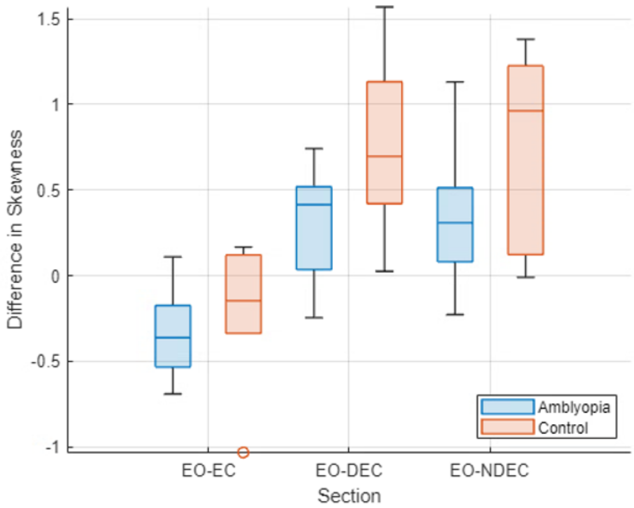

Figure 27 Boxplot with Amblyopia and Control skewness with sections containing the difference between EO and EC, DEC and NDEC in the full spectrum (0-40Hz).

No significant differences were found between the groups.

Table 7 Skewness differences between Amblyopia and Control groups (8–12 Hz)

Differences	p-value
EO-EC	0.58
EO-DEC	0.88
EO-NDEC	0.70

Table 8 Skewness differences between Amblyopia and Control groups (0–40 Hz)

Differences	p-value
EO-EC	0.60
EO-DEC	0.08
EO-NDEC	0.15

This suggests that amblyopia may not significantly impact the way EEG skewness responds to different visual conditions in the frequency ranges studied.

4.3.4 Kurtosis

The kurtosis of the EEG signal was analysed to assess the peakedness of the spectral power distribution across different visual conditions.

Table 9 Statistical comparisons of kurtosis differences between condition pairs in the alpha band (8–12 Hz)

Comparison of Differences	p-value
EO-EC vs. EO-DEC	**0.0043**
EO-EC vs. EO-NDEC	**0.0001**
EO-DEC vs. EO-NDEC	0.48

Table 10 Statistical comparisons of kurtosis differences between condition pairs in the full spectrum (0–40 Hz)

Comparison of Differences	p-value
EO-EC vs. EO-DEC	0.16
EO-EC vs. EO-NDEC	**0.0449**
EO-DEC vs. EO-NDEC	0.16

In the alpha band, kurtosis from EO-EC is significantly different from EO-DEC and from EO-NDEC. In the full spectrum, kurtosis from EO-EC differs only from EO-NDEC.

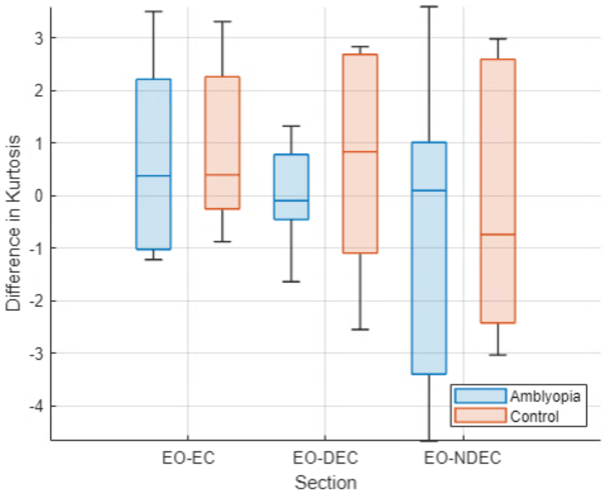

Figure 28 Boxplot with Amblyopia and Control kurtosis with sections containing the difference between EO and EC, DEC and NDEC in the alpha band (8-12Hz).

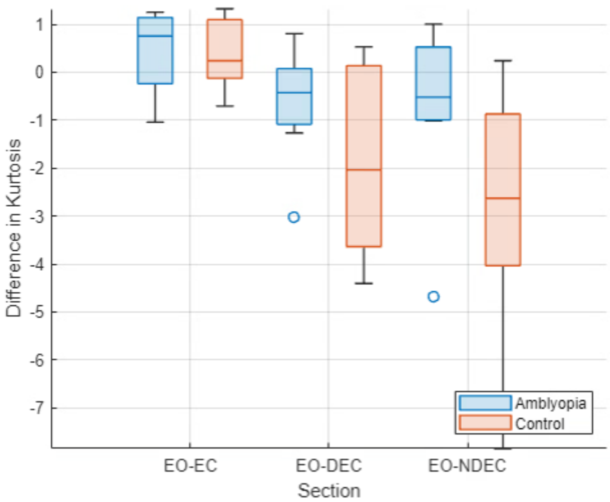

Figure 29 Boxplot with Amblyopia and Control kurtosis with sections containing the difference between EO and EC, DEC and NDEC in the full spectrum (0-40Hz).

No significant difference between groups. This suggests that for all conditions, the effect of kurtosis is similar in both amblyopia and control participants.

Table 11 Kurtosis differences between Amblyopia and Control groups (8–12 Hz)

Differences	p-value
EO-EC	0.87
EO-DEC	0.51
EO-NDEC	0.71

Table 12 Kurtosis differences between Amblyopia and Control groups (0–40 Hz)

Differences	p-value
EO-EC	0.83
EO-DEC	0.18
EO-NDEC	0.13

4.3.5 Multivariate Pattern Analysis

RF classifier was employed to distinguish between amblyopia patients and control subjects based on the extracted features of mean alpha power (in dB) and alpha bandwidth (defined as the difference between the upper and lower -3 dB cutoff frequencies). Table 13 summarizes the classification accuracies for different electrodes, conditions, and epoch lengths.

Table 13 Classification results of Mean Alpha Power (dB) at 8-12 Hz vs. Alpha Bandwidth (-3 dB) in Hz

Electrode	Condition	Accuracy (One 8s-epoch)	Accuracy (Two 4s-epoch)	Accuracy (Four 2s-epoch)
O1	EC	0.53	0.54	0.40
O1	EO	0.50	**0.70**	0.48
O1	DEC	0.67	0.68	0.39
O1	NDEC	**0.69**	0.33	**0.72**
Oz	EC	0.54	0.46	0.43
Oz	EO	0.50	0.22	0.38
Oz	DEC	0.30	0.27	0.35
Oz	NDEC	0.67	0.37	0.11
O2	EC	0.54	0.65	0.57
O2	EO	0.67	0.50	0.62
O2	DEC	0.63	0.53	0.65
O2	NDEC	0.40	0.16	0.25

The highest classification accuracy of 0.72 was achieved at electrode O1 under the NDEC condition using four 2-second epochs. Notably, the EO condition at electrode O1 also yielded a high accuracy of 0.70 with two 4-second epochs. Additionally, a notable accuracy of 0.69 was achieved at electrode O1 under the EC condition with one 8-second epoch. In contrast, the Oz electrode generally showed lower classification accuracies across conditions and epoch lengths.

The classifier's performance was further evaluated using mean alpha power (8–12 Hz) and mean theta power (4–8 Hz) as features. Table 14 presents the classification accuracies for this feature set.

Table 14 Classification results of Mean Alpha Power (dB) at 8-12 Hz vs. Mean Theta Power (dB) at 4-8 Hz

Electrode	Condition	Accuracy (One 8s-epoch)	Accuracy (Two 4s-epoch)	Accuracy (Four 2s-epoch)
O1	EC	0.69	0.61	0.46
O1	EO	0.38	0.34	0.54
O1	DEC	0.53	0.54	0.44
O1	NDEC	0.30	0.11	0.46
Oz	EC	0.46	0.58	0.44

Oz	EO	0.61	0.34	0.40
Oz	DEC	0.38	0.38	0.42
Oz	NDEC	0.61	0.34	0.42
O2	EC	0.53	0.23	**0.67**
O2	EO	**0.92**	**0.65**	0.65
O2	DEC	0.38	0.50	0.48
O2	NDEC	0.53	0.27	0.44

A good classification accuracy of 0.92 was observed at electrode O2 under the EO condition using one 8-second epoch. These results suggest that combining mean alpha and theta power can enhance classification performance, particularly with longer epoch durations.

The combination of mean alpha power (8–12 Hz) and mean beta power (16–30 Hz) was also tested.

Table 15 Classification results of Mean Alpha Power (dB) at 8-12 Hz vs. Mean Beta Power (dB) at 16-30 Hz

Electrode	Condition	Accuracy (One 8s-epoch)	Accuracy (Two 4s-epoch)	Accuracy (Four 2s-epoch)
O1	EC	**0.61**	0.54	0.44
O1	EO	0.46	0.46	0.42
O1	DEC	0.23	0.34	0.25
O1	NDEC	0.38	0.19	0.34
Oz	EC	0.53	0.50	0.40
Oz	EO	0.46	0.07	0.30
Oz	DEC	0.23	0.26	0.40
Oz	NDEC	**0.61**	0.34	0.40
O2	EC	**0.61**	0.53	0.46
O2	EO	0.38	0.42	0.36
O2	DEC	0.38	0.50	0.46
O2	NDEC	0.46	0.27	0.28

Overall, the classification accuracies using mean beta power alongside mean alpha power were moderate, with the highest accuracy being 0.61 at electrodes O1 and O2 under the EC condition with one 8-second epoch. The results indicate limited improvement in classification performance when incorporating beta power.

Further analysis was conducted using mean alpha power (8–12 Hz) and mean high-gamma power (36–40 Hz) as features. Table 16 summarises the classification accuracies.

An accuracy of 0.92 was again achieved at electrode O2 under the EO condition using one 8-second epoch, indicating that high-gamma power contributes to the classification performance. Figure 30 illustrates the scatter plots of mean alpha versus high-gamma power for the EO condition at electrode O2 across different epoch lengths.

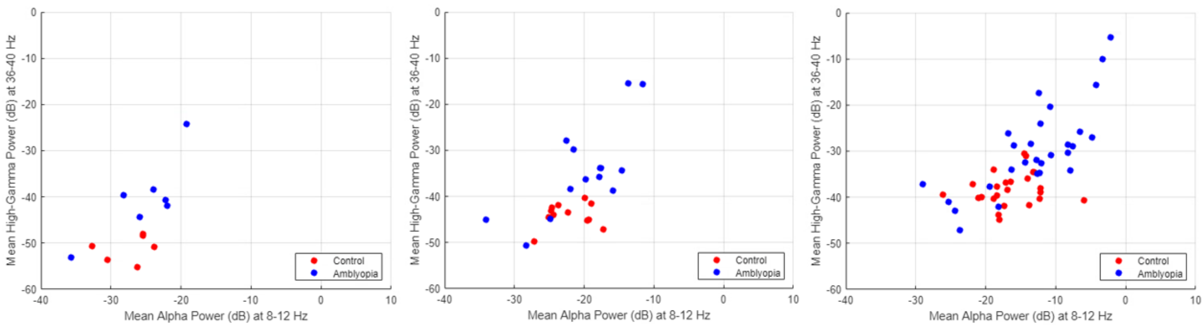

Figure 30 Mean Alpha vs High-Gamma Power for EO (First 2s removed) from O2 electrode. Left: One non-overlapping 8-second epoch. Middle: Two non-overlapping 4-second epoch. Right: Four non-overlapping 2-second epoch.

Table 16 Classification results of Mean Alpha Power (dB) at 8-12 Hz vs. Mean High-Gamma Power (dB) at 36-40 Hz

Electrode	Condition	Accuracy (One 8s-epoch)	Accuracy (Two 4s-epoch)	Accuracy (Four 2s-epoch)
O1	EC	0.61	0.61	0.52
O1	EO	0.69	0.42	0.50
O1	DEC	0.31	0.50	0.40
O1	NDEC	0.23	0.34	0.34
Oz	EC	0.31	0.65	0.61
Oz	EO	0.46	0.27	0.27
Oz	DEC	0.46	0.46	0.42
Oz	NDEC	0.53	0.50	0.42
O2	EC	0.69	0.65	0.65
O2	EO	**0.92**	**0.80**	**0.77**
O2	DEC	0.46	0.42	0.48
O2	NDEC	0.46	0.42	0.48

The classification accuracies remained relatively high at electrode O2 under the EC condition, with accuracies of 0.69 and 0.65 for one 8-second epoch and two 4-second epochs, respectively. This suggests that the inclusion of high-gamma power enhances the discrimination between amblyopia patients and control subjects, particularly at electrode O2 during the EO condition.

Across all feature sets and conditions, the impact of epoch length on classification accuracy was variable. Longer epochs (one 8-second epoch) tended to yield higher accuracies in certain conditions, such as the EO condition at electrode O2. However, there were instances where shorter epochs improved performance, as seen with four 2-second epochs at electrode O1 under the NDEC condition in Table 13.

4.3.6 Feature Matching

The highest accuracy of 83% was achieved using the feature combination Oz_DEC (Oz electrode during Dominant Eye Closed condition) and O1_NDEC (O1 electrode during Non-Dominant Eye Closed condition). This combination also resulted in a MCC of 0.65, indicating a strong positive relationship between the predicted and actual class labels. The sensitivity and specificity for this model were 0.86 and 0.79, respectively, indicating a strong ability to correctly identify both amblyopia patients and control subjects.

The top-performing feature pairs and their associated classification metrics are summarised in the table below:

Table 17 Classification results using Random Forest for feature matching

Accuracy	MCC	Sensitivity	Specificity	Feature 1	Feature 2
0.83	**0.65**	**0.86**	**0.79**	Oz_DEC	O1_NDEC
0.73	0.46	0.75	0.71	O2_EO	O1_NDEC
0.71	0.42	0.71	0.71	Oz_EO	O1_NDEC
0.69	0.39	0.64	0.75	O1_DEC	Oz_DEC

To assess the statistical significance of the classifier's performance, we conducted permutation tests with a mean accuracy of 50% and a 95% confidence interval ranging from 19.2% to 66.3%. The mean MCC from the permutation test was 0.0, with a confidence interval of [-0.61, 0.33]. The classifier's accuracy and MCC significantly exceed those obtained from the permutation test, suggesting that the observed performance is not due to random chance.

The study demonstrates that specific EEG amplitude ratios, particularly from occipital electrodes under certain visual conditions, are effective features for classifying amblyopia patients versus control subjects.

5 Discussion

This study demonstrates that EEG frequency-domain features, specifically the "fft_coefficient" features, can effectively distinguish between amblyopia patients and control subjects. Using those features, the XGB classifier achieved the highest accuracy of 85%, highlighting its capability to handle complex EEG data. The consistent values across precision, recall, and F1-score indicate a balanced performance in correctly identifying both groups. Moreover, the combination of amplitude ratio features from the Oz electrode with DEC and the O1 electrode with NDEC yielded a classification accuracy of 83% with a MCC of 0.65. This suggests that neural activity captured at the occipital area and at specific conditions is particularly sensitive to the neurophysiological differences associated with amblyopia.

5.1 Feature Analysis and Classifier Performance

We initially hypothesized that alpha activity would be present in patients with amblyopia when their fellow eye was closed, based on the understanding that alpha oscillations are associated with reduced visual input and a relaxed cortical state. Closing the fellow eye in individuals with amblyopia could lead to a state of reduced visual engagement similar to when both eyes are closed, potentially resulting in increased alpha activity. However, our results did not support this hypothesis. We did not find a significant presence of alpha activity under these conditions. This suggests that the anticipated increase in alpha power may not manifest in the same way in amblyopic patients.

While alpha band activity (8–12 Hz) is typically more pronounced during EC condition, our analysis revealed that "fft_coefficient" features were effective in classifying other conditions as well. It is hypothesised that amblyopia may induce changes in neural oscillatory activity extending beyond the alpha band or the EC condition. Supporting evidence indicates that amblyopia affects visual processing at multiple levels of the visual cortex, leading to alterations in neural synchronization [52]. Popov [53] further explains that gamma synchronisation reflects feed-forward interactions within the visual cortical hierarchy. Since the "fft_coefficient" features capture the amplitude and phase information of the EEG signal across a wide range of frequencies, differences in spectral content may contribute to the classifier's ability to distinguish between groups in EO, DEC, and NDEC conditions.

The relative scarcity of time-series features among the important predictors could be attributed to several factors. Time-series features may exhibit higher inter-individual variability [54], diminishing their effectiveness in classification tasks because patterns learned from one individual's EEG data may not generalize well to others. Additionally, these features often rely on assumptions about signal stationarity and may be particularly sensitive to noise and artifacts [55]. Given that EEG signals are inherently non-stationary, the variability within and between subjects can further reduce the consistency of time-series features across samples.

5.2 Validation Through Permutation Test

The results of the permutation test further substantiate the legitimacy of our findings. By randomising the labels and effectively breaking any true associations between the EEG features and participant groups, we observed that the classifiers' performance dropped to levels consistent with random guessing. This outcome is evident in the performance metrics, where all classifiers achieved accuracies around 50%. Such results are expected under the null hypothesis and indicate that, without genuine relationships between the data and labels, the models cannot effectively distinguish between control and amblyopic participants. The successful implementation of the permutation test serves as a crucial validation step, reinforcing that the original models were capturing meaningful patterns rather than overfitting or capitalizing on random chance.

5.3 Limitations Due to Artifacts and Data Size

An important consideration in our analysis is the potential influence of artifacts from preceding conditions on the EEG recordings of subsequent conditions. Specifically, we observed that in conditions such as EO, residual alpha activity in the 8–12 Hz band appeared in the spectrograms, likely carried over from the prior EC condition. This carry-over effect can introduce bias into the reactivity calculations, as the alpha waves present during EC may artificially inflate the alpha power observed in the following condition. Before creating Figure 24, which previously did not exclude the first 2 seconds of data, correlations between EO-EC and EO-DEC were evident; however, these correlations may be attributed to lingering artifacts rather than genuine neural responses to the current condition. This observation underscores the importance of implementing appropriate data preprocessing steps, such as discarding initial segments of EEG recordings when transitioning between conditions, to minimize the impact of such artifacts. Future studies should account for this by ensuring adequate separation between conditions or by applying more rigorous artifact correction methods to enhance the validity of reactivity and other measurements.

The limited dataset imposes constraints on the generalizability of our findings. A larger dataset would enhance the statistical power of the study, reduce the impact of outliers, and allow for more sophisticated modeling techniques that could capture complex patterns in the EEG data. Despite the small sample size, the consistency of the initial high accuracies across different classifiers suggests that the models were learning significant features relevant to amblyopia detection.

5.4 Implications and Future Directions

Regarding the multivariate pattern analysis, the combination of alpha and high-gamma frequency bands could reflect functional alterations in the visual cortex of amblyopia patients. However, these findings must be interpreted with caution. Given the close proximity of electrodes O1, Oz, and O2, one would expect similar patterns of neural activity across these sites. The lack of consistent high classification accuracies at electrodes O1 and Oz raises questions about the reliability and generalizability of the results. The discrepancy suggests that the observed high accuracy at O2 may be due to chance or uncontrolled variables rather than a robust indicator of amblyopia-related neural differences.

Furthermore, the prominence of features from O1 and Oz electrodes under contrasting conditions (DEC and NDEC) raises questions about the underlying mechanisms contributing to their discriminative power. The fact that one feature with DEC while the other with NDEC might indicate that the classifier is capturing state-dependent neural activities influenced by external factors such as variability in participant attention, alertness, or other uncontrolled variables during EEG recording. Therefore, while the initial findings are promising, further investigation is required to validate the specificity of these features to amblyopia.

Given the pilot nature of this study and the remarkable potential of these findings, it is recommended that further research be conducted to validate and expand upon these results. The high classification accuracy achieved by the XGB classifier highlights the promising role of ML techniques in analysing EEG data for amblyopia detection. Future studies should involve larger and more diverse participant populations, encompassing diverse demographics and varying severities of amblyopia, to enhance the generalizability of the results. Additionally, the university is encouraged to pursue further investigations into this promising area, as it holds significant potential for developing reliable, non-invasive diagnostic tools for amblyopia based on EEG biomarkers. Continued research could explore additional EEG features, incorporate more electrodes, and refine classification algorithms to fully harness the diagnostic capabilities revealed in this preliminary study.

6 Conclusion and Future Works

This study successfully demonstrated that EEG frequency-domain features, particularly the "fft_coefficient" metrics, are effective in distinguishing between amblyopia patients and control subjects. The XGB classifier achieved a noteworthy accuracy of 85%, underscoring its proficiency in managing the complexities inherent in EEG data.

Feature analysis revealed that while alpha band activity did not show the expected increase when the DEC in amblyopic patients, the "fft_coefficient" features remained robust across various conditions. This indicates that amblyopia may affect neural oscillatory activity beyond the alpha band and across different eye conditions. Additionally, the combination of amplitude ratio features from specific electrodes, such as Oz with DEC and O1 with NDEC, achieved a classification accuracy of 83% with an MCC of 0.65, highlighting the sensitivity of neural activity in the occipital region to amblyopia-related differences.

The permutation test reinforced the validity of our findings by demonstrating that the classifiers' performance significantly declined to chance levels when labels were randomized, confirming that the original models captured meaningful patterns rather than random noise. However, the study faced limitations related to potential artifacts from preceding conditions and a limited dataset size, which may affect the generalizability of the results. Despite these constraints, the consistency of high accuracies across different classifiers suggests that the models effectively identified significant features relevant to amblyopia detection.

Overall, the findings indicate that ML techniques, such as RF and XGB classifier, hold substantial promise for analysing EEG data to develop reliable, non-invasive diagnostic tools for amblyopia. Future research with larger and more diverse populations, improved data preprocessing methods, and expanded feature sets will be essential to validate and enhance the diagnostic capabilities observed in this preliminary study.

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Appendix A Code Availability

The code utilised in this study is publicly available in a GitHub repository. The repository is organised into separate folders for MATLAB and Python.

GitHub Repository: [https://github.com/GabrielWerpel/amblyopia-eeg-analysis]

The repository is licensed under the MIT License, permitting reuse and modification with appropriate attribution.

Data Availability:

For privacy and ethical reasons, the participant EEG data used in this study are not publicly available. Access to the data is restricted to protect the confidentiality of the participants. Requests for data access can be directed to the corresponding author and will be considered on a case-by-case basis in accordance with applicable data protection regulations and ethical guidelines.

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Research Project Coversheet and Consent Form.

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ID Number	989055637
Title of Research Project	Is amblyopia reflected in spontaneous EEG alpha rhythms?
Course Code (e.g., ENGGEN 794)	MECHENG 795A/B
Date Submitted	10/11/2024

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